Fiscal News and the Macroeconomy: Evidence from UK Budget Announcements*

Wolfram Horn †
Frankfurt School of Finance & Management

Thomas Walsh ‡
University of Glasgow

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Abstract

This paper identifies fiscal policy news shocks using high-frequency movements in UK government bond yields around Budget announcements, and traces their transmission to the macroeconomy through a local projections framework. A positive fiscal news shock (a market-assessed deterioration in fiscal sustainability) is stagflationary: output and investment contract persistently, Sterling depreciates on impact, and consumer price inflation rises in the near term. The inflationary response operates through import prices rather than domestic demand: goods prices rise sharply and persistently while services prices are essentially unchanged, consistent with rapid exchange rate pass-through. The broad-based rise in breakeven inflation across short, medium, and long maturities points beyond a pure exchange rate story toward partial fiscal dominance, with markets repricing the long-run inflation outlook in response to deteriorating debt sustainability. These findings are robust to alternative sample windows, shock definitions, and identification schemes.

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†Frankfurt School of Finance & Management, c.horn@fs.de

‡University of Glasgow, thomas.walsh@gla.ac.uk

1. Introduction

How does the economy react to news about fiscal policy? The "Mini Budget" crisis in the United Kingdom in September 2022 highlights how signals from government ministers about the future path of fiscal policy, conveyed through speeches and announcements, can profoundly affect financial markets. This crisis led to selloffs in British stocks, government bonds and currency, delivering substantially tighter financial conditions in the form of higher yields. Despite the clear impact on financial markets, there remains limited evidence of how such signals from the government impact the broader macroeconomy and the transmission channels involved, whether through financial variables, investor fears of debt monetisation, or simple devaluation effects.

Historically, addressing this question has been challenging due to issues of reverse causality and anticipation effects ahead of policy implementation. The existing literature has frequently relied on various identification strategies: narrative selection based on historical record, military spending news, forecast revisions, and structural VAR approaches.

Our identification of fiscal news shocks instead relies on high-frequency reactions of financial markets to periodic and comprehensive policy announcements by the UK government in Budget Speeches to the UK Parliament. These regular fiscal events cover the full range of changes to government spending, taxation, and borrowing. They are televised on all mainstream channels and highly salient in domestic news media.

We isolate fiscal news shocks by measuring the high-frequency reactions of UK government bond yields along the yield curve, capturing the potentially heterogeneous impact on short and long yields. Reverse causality from macro news to fiscal policy variables can be ruled out due to the tight reaction window around the fiscal events. The core idea here, similar to that guiding the use of forecast revisions, is that financial markets continuously update expectations based on all available information, meaning any dynamics just after the fiscal event reflect newly received fiscal

information and not confounding macro news.

Our results show gilt yields rise on impact and remain persistently elevated over the first 100 trading days following a fiscal news shock, consistent with markets revising upward their assessment of future borrowing costs and fiscal risk premia. Equity prices and sterling fall contemporaneously, while breakeven inflation (based on the spread between nominal and inflation-protected bonds) responds significantly across the maturity curve, pointing to an upward revision in inflation expectations alongside the deterioration in perceived fiscal sustainability.

Turning to macro outcomes at lower frequency, fiscal news shocks are strongly contractionary in output — both GDP at quarterly frequency and industrial production at monthly frequency fall persistently and significantly. Labour market conditions deteriorate in tandem. Price responses are inflationary up to medium horizons, and goods-sector CPI inflation displays clear stagflationary behaviour, reflecting the pass-through from sterling's depreciation into import prices. Services inflation, driven mostly by domestic wage pressure, does not respond. The trade balance improves with depreciation, but this hides early contractions in both imports and exports, which display a clear J-curve effect.

The fiscal situation also deteriorates through lower tax receipts and elevated expenditures at medium horizons as automatic stabilisers activate. To some degree, the initial loss of market confidence is vindicated by subsequent fiscal outturns, giving a distinct self-fulfilling character to our results. A fiscal news shock which increases the 1Y gilt yield by 25bpts, the standard scaling in the monetary policy literature, is associated with increases in the deficit of 2 percentage points at peak, and with a Debt-to-GDP ratio 10 percentage points higher than baseline 5 years after impact.

We show that the Bank of England (BoE) has been historically slow to offset the tightening in financial conditions, with BoE policy rate ("Bank Rate") not easing until well into the second year after the shock. This is consistent with the BoE facing a stagflationary trade-off created by the adverse fiscal news shock.

Our paper makes three contributions. First, we construct the first high-frequency measure of fiscal policy news shocks for the UK, exploiting the institutional features of Budget announcements.

Second, we document a transmission mechanism that is fundamentally distinct from monetary policy transmission: fiscal news shocks generate near-immediate inflationary pressure through exchange rate pass-through to import prices, rather than the slow wage-cost channel through which monetary shocks reach the price level. The differential response of goods and services CPI provides direct evidence on this channel in a way that aggregate inflation measures cannot.

Third, we show that the broad-based rise in breakeven inflation across maturities is inconsistent with a pure exchange rate story and points toward partial fiscal dominance, or at least the fear of partial monetisation: markets appear to reprice long-run inflation expectations in response to perceived deterioration in debt sustainability, even absent any explicit change in monetary policy. This channel is largely absent from the existing UK fiscal literature, which has focused on tax multipliers and spending effects rather than the financial market and inflation expectations dimensions of fiscal news.

Our work is therefore complementary to existing analysis, expanding the evidence base for fiscal transmission in an open economy setting, where the exchange rate and import price channels are first-order in a way they are not in US-focused work.

Our main results are robust to a wide range of robustness checks across alternative regression estimation windows, choice of maturity of the fiscal surprise, alternative shock constructions and identification schemes, and we present evidence against government information channels and instead reinforce our interpretation of the shocks as reflecting fiscal sustainability fears.

The rest of the paper is structured as follows. The following Section sets out related work and situates our contribution therein. Section 4 describes the institutional setting relating to UK fiscal policy and Budget announcements. Section 5 details our empirical strategy and how we identify

fiscal news shocks. Section 6 presents our results and 7 examines their robustness to changes in identifying assumptions. Section 8 concludes the paper.

2. Related Literature and Contribution

This paper relates to a broad literature that investigates the macro effects of fiscal policy. Historically, this has been a difficult task. Government choices regarding taxation, spending, debt issuance, and debt management impact macro variables, but are also set with these variables incorporated into decision making. Most simply, fiscal identification faces the dual hurdles of simultaneity and anticipation effects. The key methodological innovation of this paper is to import methods of the monetary policy literature (high frequency identification) and use the beneficial properties of this identification scheme to overcome these historical challenges.

The literature has developed several tools to address endogeneity issues. The first broad class involves structural identification inside vector autoregressions (Blanchard and Perotti 2002, Mountford and Uhlig 2009). While tractable, SVAR approaches are vulnerable to the anticipation critique – if agents observe fiscal news before it is implemented, VAR residuals will not recover the true structural shock (Leeper et al. 2013).

The narrative approach has tried to overcome anticipation effects by directly interpreting historical records to identify exogenous changes in fiscal policy based on stated motivations of policymakers at the time, such as, but not limited to, military spending and warfighting concerns (Ramey and Shapiro 1998, Ramey 2011, Romer and Romer 2010). Historical records and news media-based shocks find that tax increases are strongly contractionary.

Cloyne 2013 applies the Romer-Romer approach to postwar tax changes in the UK, finding a 1 percentage point cut to taxes raises GDP by approximately 2.5 percent in the medium term. This provides a close benchmark for comparison to our paper. However, narrative approaches

rely on realised changes in taxation and spending rather than the information content of fiscal announcements. It is possible the government could announce a suite of future policies, and then radically alter or halt plans based on public and market reaction to the news in the intervening period (Leeper et al. 2013; Schmitt-Grohé and Uribe 2012). In this sense no policy has changed, yet there was news when the plans were announced. To tackle the issue of news directly Ben Zeev and Pappa 2017 construct fiscal news shocks as orthogonal to current fiscal policy but predictive of further developments in the medium run (a max forecast error variance, max-FEV, estimator of Barsky and Sims 2011 used to identify TFP news) .

High-frequency identification (HFI) used in monetary policy analysis (such as Gürkaynak et al. 2005) is most closely related to our approach. Surprises are identified as movements in financial instruments in a narrow window around policy announcements Kuttner 2001; Gürkaynak et al. 2005; Nakamura and Steinsson 2018. The key identifying assumption is that within this window, asset price movements reflect only the news content of the policy decision. Bu et al. 2021 extend high-frequency identification across the full yield curve, constructing a unified measure of monetary policy shocks that captures movements at all maturities simultaneously. Miranda-Agrippino 2016 refine HFI further by purging monetary surprises of an information effect: the component reflecting the central bank's private information about the economic outlook rather than a genuine policy change.

The application of high-frequency identification to fiscal policy is more recent and comparatively sparse. Born et al. 2013 study the effects of fiscal announcements on financial markets in a cross-country panel. Kerssenfischer and Schmeling 2024 identify incoming macro news, data prints and policy briefs, and high-frequency market reactions. Bi et al. 2025 and Lengyel 2022 construct debt news shocks for the US and UK respectively using surprises in bond issuance, finding significant effects on financial conditions and real activity. Our identification via yield movements around Budget announcements is complementary – where Bi et al. 2025 and Lengyel 2022 ex-

ploit quantity surprises in debt issuance, we exploit price surprises in the gilt market around fiscal announcements.

A complementary approach uses professional forecaster data to construct fiscal news. Ricco et al. 2016 construct fiscal news shocks using Survey of Professional Forecasters (SPF) forecast revisions for the paths of federal government consumption and investment, showing that clear policy communication can coordinate expectations and that fiscal multipliers are amplified when forecaster disagreement is low. Our yield-based measure is related but distinct: gilt market movements capture both the direct fiscal impulse and its implications for debt sustainability and future monetary policy, and are available in real time rather than at survey frequency. Ramey 2011 and Leeper et al. 2013 emphasise more broadly that correctly dating when fiscal news arrives — rather than when policy is implemented — is crucial for recovering unbiased impulse responses.

The transmission of fiscal news through financial markets connects our paper to the literature on fiscal sustainability and the yield channel. Laubach 2009 and Engen and Hubbard 2004 document the empirical relationship between fiscal deficits and long-term interest rates. Corsetti et al. 2013 show that fiscal multipliers depend critically on perceived debt sustainability — when debt is elevated, fiscal expansions may prove contractionary if they raise sovereign risk premia and crowd out private investment. The response of gilt yields to fiscal announcements in our data is consistent with this channel, and we use the term spread decomposition to disentangle demand effects from sustainability effects.

3. Stylised Model Framework

This section lays out the components of a more detailed model through which we would expect to see fiscal news shocks transmit to macroeconomic variables.

The environment is a small open economy populated by households, firms, and a government.

There are three sectors: non-traded (services), traded (goods), and an international bond market. Households are standard expected utility maximisers, consuming $C^T + C^N$.

Firms in the traded sector produce with an imported intermediates share α, price with some stickiness, and face a (linearised) marginal cost:

$$mc_t = \alpha_M(p_t + e_t) + (1 - \alpha_M)w_t \tag{1}$$

The government taxes, spends, issues nominal bonds B_t, and runs primary surpluses s_t. The nominal government debt stock evolves according to the standard law of motion:

$$B_t = (1 + i_{t-1})B_{t-1} - P_t \cdot surplus_t \tag{2}$$

implying an intertemporal debt-solvency condition (meaning in the long run the government can meet its nominal obligations). The condition has two components: the value of expected surpluses and the seignorage through an expected inflation component[1].

$$\frac{B_t}{P_t} = \beta\mathbb{E}_t \sum_{j=0}^{\infty} \beta^j surplus_{t+1+j} + \beta\mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left(\pi_{t+1+j} \frac{B_{t+j}}{P_{t+j}}\right) \tag{3}$$

We assume the international bond market prices UK government debt according to:

$$i_t = i^* + \mathbb{E}_t \Delta e_{t+1} + \phi_B \cdot B_t/Y_t \tag{4}$$

where B_t/Y_t is the debt-to-GDP ratio, the key measure of debt sustainability, and $\phi_b > 0$ is the sovereign risk premium elasticity, consistent with preferred-habitat and safe-asset demand frameworks (Greenwood and Vayanos 2014; Krishnamurthy and Vissing-Jorgensen 2012). A fiscal

1. The version without a Fisher approximation: $\frac{B_{t-1}(1+i_{t-1})}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} \left[\frac{\prod_{k=1}^{j}(1+\pi_{t+k})}{\prod_{k=0}^{j-1}(1+i_{t+k})}\right] surplus_{t+j}$

news shock which raises the expected path of debt-to-GDP raises the risk compensation the market demands on UK government bonds. The nominal yield and depreciation of Sterling will adjust accordingly.

One can think of a fiscal news shock as an unanticipated revision to the expected path of primary surpluses:

$$\xi_t = \mathbb{E}_t \sum_{j=0} \beta^j surplus_{t+j} - \mathbb{E}_{t-1} \sum_{j=0} \beta^j surplus_{t+j} \tag{5}$$

A negative shock, $\xi_t < 0$, a fiscal deterioration, triggers three simultaneous responses:

Channel 1: Financial conditions Gilt yields rise as markets price higher expected debt.

$$\Delta i_t = \phi_B \Delta b_t$$

where $b_t = B_t/Y_t$. This triggers standard IS-curve-type spending reduction from households and investment reduction by firms due to elevated user cost of capital.

Channel 2: Fiscal Dominance and Monetisation Risk Expected inflation also reacts to the fiscal deterioration ξ through the fear of debt monetisation.

$$\Delta \pi_t^e = \theta \cdot \xi_t$$

Here the parameter θ reflects the degree of fiscal dominance (Sims 1994; Woodford 2001; Sargent and Wallace 1984). In the Ricardian regime, $\theta = 0$ and expectations are unaffected. Monetary policy is dominant and sets the short term policy rate, and UIP determines the exchange rate. In the seignorage regime of fiscal dominance $\theta > 0$ and fiscal shocks shift inflation expectations. We find evidence of this channel through breakeven inflation measures derived from spreads between nominal and inflation-protected bonds.

Channel 3: Depreciation and Imported Inflation The currency devalues by standard channels

$$\Delta e = f(\phi_B \Delta b_t^e)$$

and prices adjust differentially based on import exposure:

$$\Delta \pi_t = \alpha \Delta e_t + (1 - \alpha) \Delta w_t$$

The stylised model makes several predictions and also helps us discriminate between channels.

The third prediction is key to discriminating between a pure exchange rate pass-through story, Channel 3 only, which predicts that long-run inflation expectations can remain anchored even in the face of short-run inflation. This predicts that any rise in breakeven inflation should affect only the short-end of the inflation curve and gradually die out at longer horizons. Partial Fiscal dominance, however, would predict a rise across the inflation curve.

Prediction	Channel
Output and investment contract	1
Gilt yields rise, sterling depreciates	1,3
Breakeven inflation rises uniformly across maturities	2
Tradable goods inflation rises, services determined by domestic wages	3

Table 1: Stylised Model Predictions Testable in the Data

4. Identification

4.1 Institutional Setting - Budget Announcements

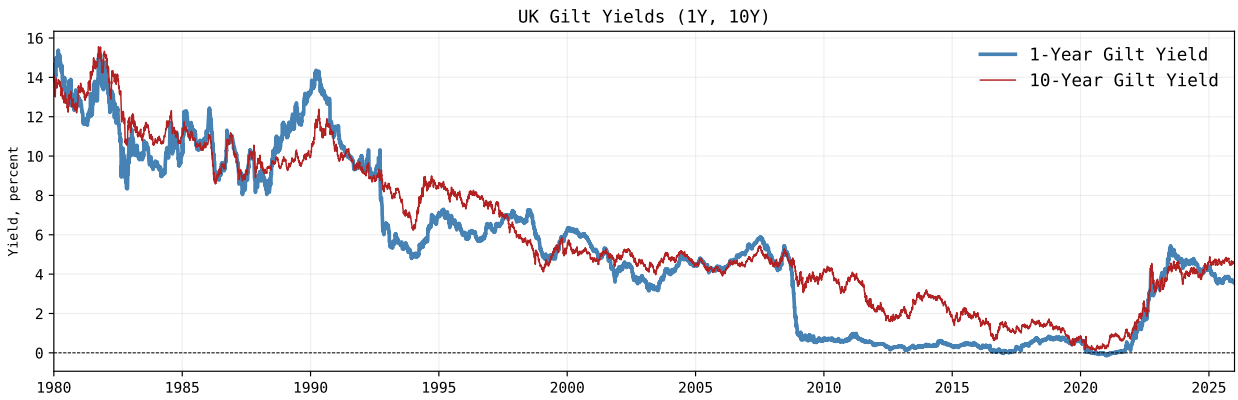
The regular and comprehensive announcements of the UK government's plans for spending, taxation and debt management is referred to as The Budget. The Budget is presented to the elected lower house of the UK parliament, the House of Commons, by the equivalent of the Finance minister (*Chancellor of the Exchequer*) once a year in the Spring. Since 1997 there have been both Spring (usually March to coincide with the new fiscal year starting in April) and Autumn (September to November depending on year) Budget releases, and some emergency Budgets, for example in 2010 during the global financial crisis, after a change of government. The time series of budgetary events is necessarily sparse.

Budgets are broad in scope and highly salient, receiving significant media coverage. Photos of the Chancellor posing with the traditional red leather briefcase, supposed to contain the speech, are common in domestic media. The typical speech in recent decades is around one hour to one and a half hours in length, and was longer in the past (confirmed by House of Commons Library timestamps and also speech text word counts).

However this is only the initial outline of policy changes, after which the leader of the opposition responds, followed by days of debate, usually four in total, each with a specific theme. Since 2012 every Budget event has been followed immediately by an assessment from the independent fiscal forecaster the Office for Budget Responsibility (OBR).

4.2 Gilt Yield Surprises as Fiscal News

We quantify incoming fiscal news by examining the reaction of fast-moving financial markets around these fiscal events. Built into this methodology are the two assumptions: First, that prior to the event, all market relevant information is "priced in" by markets. Jumps in financial variables



Note: the above figure presents daily yields for the 1 and 10 year UK government bonds.

Figure 1: UK Government Bond Yields (%)

around the time of fiscal announcements are thus attributable to market revisions following new fiscal information; and second, that the window is small enough so that markets do not update based on other incoming news.

We define the fiscal news surprise at a particular maturity along the yield curve, m as the change in yields, in a short window $[0, +\Delta]$ around the time of the announcement τ on day d

$$s_d^m = \begin{cases} (\text{Yield}_{d,\tau+\Delta}^m - \text{Yield}_{d,\tau}^m) & \textit{on announcement days} \\ 0 & \textit{otherwise} \end{cases} \tag{6}$$

The size of the measured event window Δ is a key parameter in high-frequency identification. A short window (e.g. 10 minutes) ensures that confounding news arriving is almost certainly ruled out, but this comes at the cost of potentially not capturing the full response which may take longer to materialise as market participants digest new information. Expanding the event window thus creates the tradeoff between capturing the full dynamic response and introducing potentially confounding news. We take a balanced approach and use daily changes. This is likely to give markets enough time to update and analyse Budget speech contents thoroughly, while also limiting

exposure to noise.

Fiscal announcements in the UK are typically long, with speeches often lasting an hour or longer, and involve numerous changes to policy details. We argue that it is natural that markets may be slower to learn from announcements compared to monetary policy announcements which tend to operate through a simpler menu of choices (25 basis points changes are the norm) and far fewer policy instruments. Conversely, fiscal announcements can potentially contain a large number of instruments and continuously distributed alterations, with often complex interactions with the existing system and other announced changes.

The core delivery mechanism of UK fiscal policy differs fundamentally from Bank of England monetary policy announcements - a televised speech before publication of documents, compared to instant digital publication at 12:00 for MPC decisions. Monetary policy events take place more often, 8 times per year compared to typically twice for fiscal policy, and the BoE policy rate adjusts instantly, while fiscal policy adjustments may require some lead-in time.

4.3 Confounding Daily Macro News

To prevent jumps in yields caused by incoming macro news being falsely attributed to domestic fiscal events, we remove Budget announcements on days which coincide with central bank announcements on monetary policy decisions (Bank of England, European Central Bank, Federal Reserve). We follow Kerssenfischer and Schmeling 2024, who document which macro news moves markets across countries, institutions, and instruments.

4.4 Sample Period

Our baseline sample runs from 1980 until 2019 inclusive. The choice of this range as our default sample is motivated by four considerations. First, this is the earliest starting point which gives us access to sufficient macroeconomic time series and therefore regression degrees of freedom.

One could argue in favour of alternative starting points due to, for instance, structural shifts with globalisation and the great moderation; however, this involves sacrificing a non-trivial fraction of the overall sample.

Second, to avoid pandemic complications and otherwise unusual dynamics in 2020 and 2021. Nevertheless, results are broadly similar with or without the pandemic. These results are reported in the appendix.

Third, inference around the Truss-Kwarteng mini-budget shock of 2022 is complicated not just due to post-covid dynamics, but also the size of the September 2022 shock is an extreme outlier compared to the rest of the budget announcement surprises (equivalent to a 6SD event using variation on non-budget days).

This is potentially problematic for inference. One can decompose the overall impulse response function at horizon h estimated by OLS, β^h, as the weighted average of macro-dynamics around each individual event β_t^h where the set of events is: $\mathcal{E} := \{t : t = \text{announcement day}\}$, assuming demeaning:

$$\hat{\beta}^h = \frac{\sum_{t\in\mathcal{E}} s_t \cdot (Y_{t+h} - Y_{t-1})}{\sum_{t\in\mathcal{E}} s_t^2} = \frac{\sum_{t\in\mathcal{E}} s_t \cdot (\beta_t^h)}{\sum_{t\in\mathcal{E}} s_t^2} = \sum_{t\in\mathcal{E}} w_t \cdot \beta_t^h = [T^{-1}\sigma_S^{-2}] \cdot \sum_{t\in\mathcal{E}} s_t \cdot \beta_t^h \tag{7}$$

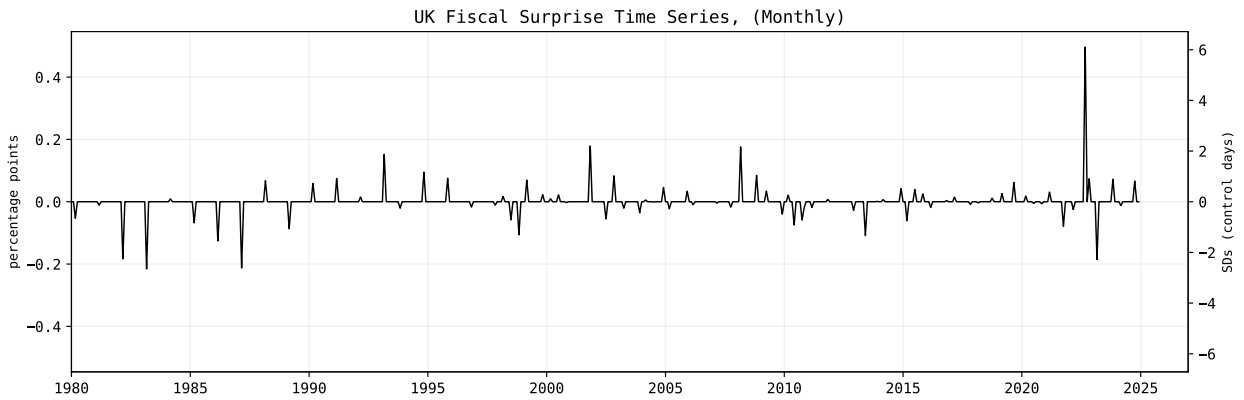
Hence our baseline sample is set conservatively to only incorporate information from fiscal events with "normal" dynamics $(Y_{t+h} - Y_{t-1})$ and avoids placing extreme weight w_t on any single event. We explore several alternatives which incorporate a fuller sample until 2026, and truncate the Truss-Kwarteng shock in two ways: setting it to zero, or setting its absolute size equal to the next largest shock in the period 1980-2019. Inclusion of the period 2020-2026 increases the persistence and flattens the profile of yield responses to fiscal news shocks.

Finally any event after the date $t - H$ will not have a full outcome data sequence, and so will contribute a so-called ragged-edge to the above aggregation unless incomplete sub-events are

excluded. We explore in our appendix how our results evolve with sample decisions related to these issues and propose some solutions.

4.5 Timeline of Fiscal Events

An informal check of our time series of shocks is to match against major national and world events. Highlighted in the series are a selection of the largest shocks and suggested interpretations.



Note: the above figure presents the fiscal news shock series for the 3 year maturity yields. The left axis measures percentage points and the right axis is in standard deviations of the interday change in 1Y yields outside of announcement days. Large positive shocks which occur close of major events have been highlighted.

Figure 2: Fiscal News Surprise time series, 1980-2025

Some have clear drivers such as the shock of Autumn 2022, or the dubbed "Black Friday" currency crisis of 1992 in which the UK government was forced out of the European Exchange Rate Mechanism (ERM), and effective peg against the West Germany Deutschmark at DM 2.95, with a tolerated range of six percent. After crashing out of the ERM, Sterling fell to below 2.40 by October 1992. One may perhaps read the persistent negative Budget surprises of the ERM period as the Government repeatedly under-delivering relative to what would be expected to defend the ERM peg.

Another major event in the fiscal surprise timeline is the first Budget after the September 2001

terrorist attacks. In October 2001, it was announced UK armed forces would participate in "Operation Veritas". November 2001 was the first fiscal event after this announcement of the military campaign in Afghanistan.

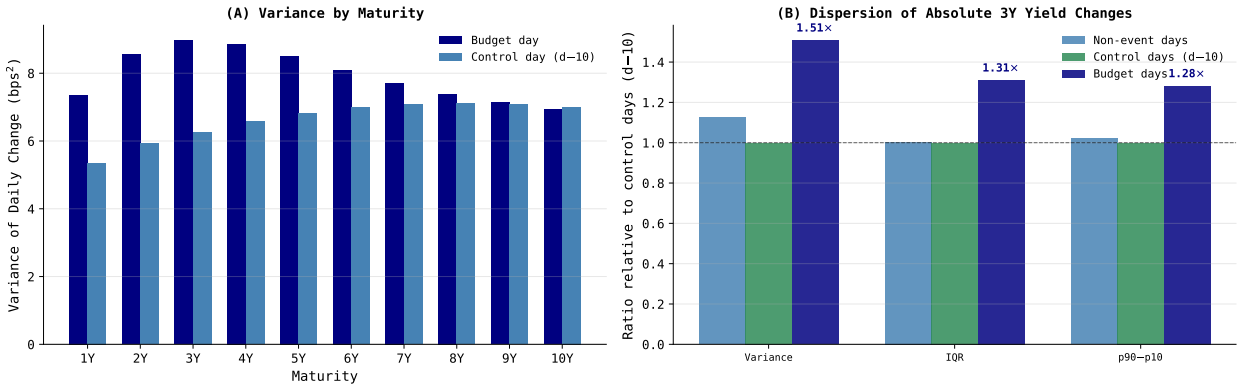
The March 2008 budget was at the peak of the Global Financial Crisis. Bear Stearns was collapsing in real time, sold to JPMorgan on 16 March 2008, just days after the Budget. In the UK, Northern Rock had been nationalised the month before in February 2008. Thus the 2008 budget was a large reorientation of the government's fiscal stance. Both events can be interpreted as government announcements of the details of major new spending commitments and alterations to the government's borrowing trajectories, even if the broad direction of travel was already assumed by markets, the exact specification of operations, and the resolution of uncertainty drive the surprise.

We therefore argue these event-day yield surprises can be broadly interpreted as debt sustainability surprises - higher yields on event days representing markets' assessment of worsening sustainability.

4.6 Variation on Budget Days versus Control Days

As presented in Figure 3, Budget days feature changes yields which are substantially more dispersed than on control days (non-announcement days, ten trading days before the fiscal event). Variance at every maturity from one to ten years is larger on event days, with the largest difference at the 3-year maturity. This confirms that Budget announcements generate movements in yields that are much larger than typical non-Budget days. For this reason, to maximise the effectiveness of our empirical identification strategy, we opt for the 3-year yield as the basis for our fiscal surprise variable. The 3-year is not substantially different from other short-end yields from 1-5 years; A principal component analysis of the full panel of yields across yields 1-16 years (for which a full panel is available) confirms a strong factor structure, with 99 percent of variance explicable by the first 3 factors (usually interpreted as "level", "slope", and "curvature" factors in the litera-

ture, based on the factor loadings). The level factor responds in a very similar way to (e.g.) the shorter-end 1-5 year yields.



Note: Blue bars represent variance of interday changes in yields on Budget announcement days. Light blue bars represent control days. Controls events are ten trading days before the fiscal event.

Figure 3: Variance of Yields on Announcement Event Days and Control Days

4.7 Aggregation across frequencies

We aggregate daily gilt yield surprises to monthly and quarterly frequency by summing within the period, so that the aggregated surprise series captures the total fiscal news content rather than its average intensity. In practice, the vast majority of months contain no Budget announcement and therefore have a surprise of exactly zero.

Aggregating to monthly frequency also allows us to incorporate macroeconomic controls — output, inflation etc. are unavailable at high frequency but are important for isolating the fiscal transmission channel. The jumps in yields can only plausibly be used to identify fiscal shocks in conjunction with the macroeconomic covariates that would fit residualised-surprises that are unpredictable. Without controls we cannot make as-good-as-random arguments.

Therefore, other fast-moving variables such as exchange rates and interest rates are aggregated by taking the within-period mean, which for step-function variables like the Bank of England base

rate produces monthly values off the grid induced by 25 basis point changes, reflecting the timing of any rate change within the period.

5. Estimation

We run the following series of local projection regressions for each horizon $h = 1$ to H after the fiscal news shock (Jordà 2005; Jordà and Taylor 2016, 2025) estimated primarily by OLS. Our outcome of interest is the long-difference in Y, normalised to the period preceding the fiscal news shock, $t - 1$. HAC standard errors with Bartlett weight $bw = (h + 1)$ are employed throughout to capture the dependence structure in regression residuals e_{t+h} or u_{t+h} due to the long-difference structure of the dependent variable, $Y_{t+h} - Y_{t-1}$.

Our daily regressions are ordered by trading day, meaning a following trading Monday is treated as the direct successor of the previous trading Friday, if markets were open on both days. Similarly, trading time skips other calendar days on which markets do not trade. At the trading-day frequency we use:

$$Y_{d+h} - Y_{d-1} = \alpha_h \cdot s_d + \mathbf{W}'_{d-1}\Omega_h + \mu_{dw(t)} + \mu_{wy(t)} + e_{t+h}; \qquad \forall h \in 1, 2, ..., H = 100 \tag{8}$$

Where we control for one-day-lagged 1Y and 10Y yields and FX rate, as well as fixed effects for day-of-the-week (dw) and week-of-the-year (wy) to capture cyclicality within the trading week, as well as slower-moving seasonality over the year.

At the monthly frequency we run a similar specification, but this version is able to incorporate macroeconomic controls.

$$Y_{t+h} - Y_{t-1} = \beta_h \cdot s_t + \mathbf{X}'_{t-1}\Omega_h + \mu_{m(t)} + u_{t+h}; \qquad \forall h \in 1, 2, ..., H = 60 \tag{9}$$

We include as controls specific lags of the fiscal surprise to remove any potential autocorrelation (and therefore predictability) of the fiscal news shocks. One particularly important concern in the high-frequency identification literature is the predictability of the surprise series with lags of macro variables. Ramey 2016 states shocks should have the property that they are orthogonal to the current information set.

This predictability or autocorrelation structure is typically interpreted in the monetary policy literature as a central bank information effect: central bank announcements contain monetary policy shocks, as well as revealing information about the central banks reaction function and superior information set. Miranda-Agrippino 2016; Miranda-Agrippino and Ricco 2021 show that filtering for the predictable component in monetary surprises generates IRFs which conform much better to economic theory, removing so-called price puzzles. The process is a double filtering, first in event-sequence an AR(1_e) to remove FOMC information effects, and then with the fitted residuals an AR(12_m) in calendar time sequence to account for potential autocorrelation due to slow updating.

Further work by Lloyd and Manuel 2024 show that a two-stage process of filtering the predictable component out of high-frequency surprises to isolate exogeneous shocks is inefficient compared to a one-stage estimation that contains all the filtering covariates, an application of the Frisch-Waugh-Lovell (FWL) partial regression theorem.

Therefore in the baseline specification we control for lags at 6, 12, 24, 48 and 60 months to absorb the periodicity fully. This mirrors Miranda-Agrippino and Ricco 2021's event-sequence filter to remove event-autocorrelations. Such correlation is due to a modest number of events in the 1980s, and is stronger in the 1-year yield than other maturities.

One potential explanation for autocorrelation in event surprises is that governments try to avoid large market disruptions (in either direction), which might generate incomplete announcement of the full set of government plans. One could call this an "unfinished business" driver of surprise autocorrelation. Since the UK government was trying to defend a currency peg (and eventually

failed dramatically under speculative assault) it is natural they may have been trying not to spook markets around any individual announcement event, but long-term plans did not alter much. Whatever remains unannounced in one event is left over for the next. This may be plausible in normal times when a government wants to minimise uncertainty around fiscal events, however it is clearly unhelpful during a crisis when markets need certainty across the full range of potential future states.

We validate our empirical approach to shock construction by fitting the fiscal news shocks as the residualised surprises (removing variation due to controls), and test for predictability of these fitted shocks as in Känzig 2021, for example. The fitted shocks cannot be predicted with lags of various macro series at monthly and quarterly frequency, and the interday surprises themselves cannot be forecasts with daily data around the time of the announcement, only previous past surprises are predictive of the next surprise.

5.1 Controls

We add 12 lags of the outcome, and control for the state of the macroeconomy by including 12 lags of a vector of macroeconomic controls typically found in the SVAR literature, proxying for output, inflation, unemployment, and interest rates. The corresponding order of the lag polynomial adjusts to 4 in the quarterly specification to preserve the one year history.

We use change in industrial production for monthly frequency LPs versus real GDP growth quarterly, as well as including the 1Y gilt yield, and the headline (year-on-year) CPI inflation rate. Seasonal dummies for months- or quarters-of-the-year are also included.

5.2 Shock sizes

We normalise the fiscal surprise series employed in the daily regressions such that IRFs report responses to 1SD(fiscal surprise). One can in parallel view this as rescaling the surprise to have a

SD of 1.

In the main macro LP regressions with controls added, we normalise the fiscal news shock s_t such that it increases the contemporaneous monthly 1Y gilt by 25 basis points. Such a normalisation is standard in the broader macro and monetary literature. This monthly series is not rebased when it is aggregated to quarterly, preserving the interpretation that a 1 unit surprise increases the monthly 1Y yield by 25 basis points (bpts) even in the quarterly LP.

This scaling is particularly notable since the short term rates continue to climb for some time after the initial shock. Note there are approximately 20 trading days in a month and 252 trading days in a year. Our daily IRFs of gilt yields to fiscal surprises reach peak response around 60 days after impact and stay elevated even at the full 100 day horizon. This suggests these shocks differ from monetary policy shocks in their persistence. Monetary shocks typically feature strong one-and-done short persistence.

It is this size and relatively high persistence which will generate large real effects. From the monetary economics literature, it is not unusual to estimate output losses with a maximum absolute response of around 3 percent to a 100 basis points shock. The 1Y yield achieves a similar sized response in the monthly LP regressions after some months, so approximate orders of magnitude are similar.

The difference between our estimates and those in the monetary literature likely reflects two factors: (i) the setting: the UK is a small open economy, whereas US shocks are potentially dampened by reserve currency demand and (ii) the greater persistence of fiscal news shocks relative to monetary shocks, with gilt yield responses lingering in the financial system for several months rather than reverting quickly.

5.3 Impulse Response Functions

We present our results to fiscal news shocks as impulse response functions up to a horizon of five years after impact. All regressions above are estimated by OLS. We use HAC standard errors to account for heteroskedasticity over time as well as serial dependence of errors, with Bartlett weights of $h + 1$, i.e at horizon 7, serial correlation up to 8 steps is allowed. Uncertainty around point estimates is shown as the shaded error bands which represent the 68 percent / one standard error (darker area) and 90 percent confidence intervals (lighter area) respectively.

6. Results

This section presents and discusses our primary empirical results. We find a positive-signed fiscal news shock, representing an increase in yields and tightening of financial conditions, is broadly stagflationary in the sense that it depresses output and labour market variables, while at the same time devaluing the currency and increasing inflation. While our work shares many similarities with work in the area of monetary analysis through high-frequency identified monetary policy shocks, this is our main point of divergence – a fiscal news shock, broadly interpreted as a fiscal sustainability shock, pushes CPI yoy inflation up, compared to a negative or null response in the MP literature. Furthermore, we show this increase in CPI inflation is driven entirely by goods, the majority of which are imported goods, and not services which depend much more on the domestic wage rate.

We separate responses by speed of variables: daily, monthly, quarterly. The next section discusses further analysis

6.1 Transmission Channels from News to Real Economy

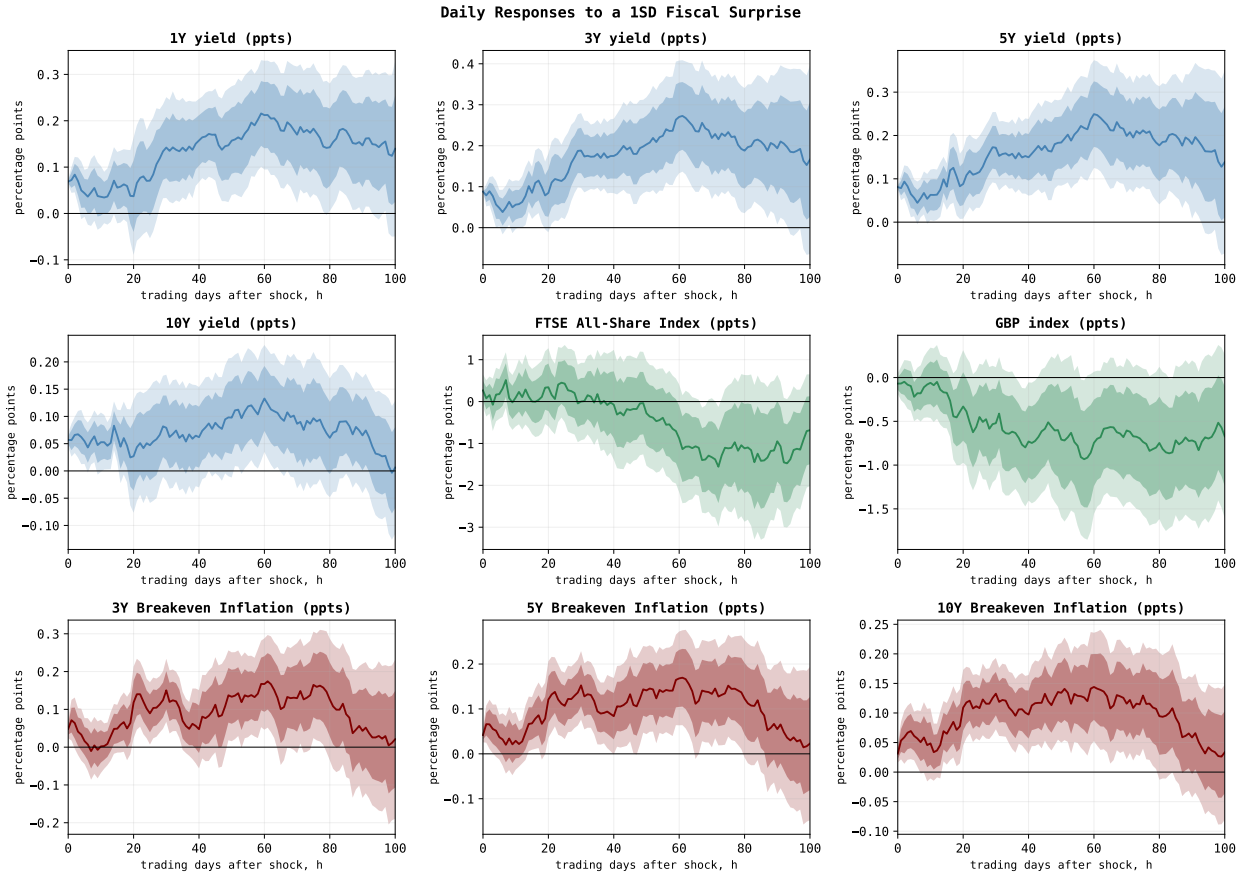
We see fiscal news shock transmission to macroeconomic variables happening through three distinct channels: (i) an interest rate channel acting through financial conditions influenced by government bond yields. Evidence for this channel consists of impulse responses to fiscal surprises as the daily and monthly frequencies (featuring macroeconomic controls) (ii) a fiscal risk channel whereby markets price in future (partial) monetisation or fiscal dominance. Evidence in favour of this channel would be the almost-uniform rise in breakeven inflation expectations based on market data across maturities seen on daily IRFs. A simple FX movement would be priced-in mostly at the very short end of the so-called inflation curve. The fact that even the 10Y-breakeven responds suggests this is about medium term expectations (iii) an imported inflation channel through FX movements and a reliance on imported goods. We provide evidence that core CPI inflation (excluding energy and food) responds significantly to fiscal news shocks, while services CPI is essentially flat over the horizon. Core CPI loads significantly on imported foreign goods, while Services CPI is driven primarily by the domestic wage rate, and therefore, the state of the domestic economy.

The interest rate channel is contractionary while the fiscal risk and the open economy channels are inflationary. This is what distinguishes UK fiscal shocks from US-focused previous literature. In a small open economy the third channel - the price of imports - is first-order, while for a large and relatively closed economy like the United States this channel will be weaker.

6.2 Daily Responses of Gilt Yields and Breakeven Inflation

Nominal interest rates respond strongly in early periods after the shock. The fiscal news shock generates a significant positive response in the short and longer maturities in initial months after impact, and there is a degree of persistence in both IRFs - reaching a maximum closer to 60 trading

days after impact. This represents a substantial financial shock and contrasts markedly to credibly-identified monetary policy shocks. Monetary shocks are characterised by very low persistence in the yield responses, especially at the short end. Typically in a monthly or quarterly VAR, yield IRFs could be summarised as "one and done" given how short the effects last. This is not the case with fiscal news, which takes several months to percolate though the financial system and achieve a peak response, before lingering with a much higher persistence than a monetary shock.



Note: Dynamic responses estimated according to the equation (8) above. Shaded areas represent 68 and 90 percent confidence intervals respectively. Standard errors are heteroskedasticity and autocorrelation-robust up to $h+1$ lags. IRFs are colour according to yields, Stocks and FX and Breakevens. The fiscal surprise is scaled to the standard deviation of the surprise series measured on event days: $\sigma[s_t;\ t \in \mathcal{E}]$

Figure 4: Daily Responses of Financial Markets to a Fiscal Surprise after Event Days

Stock prices decline, albeit gradually and with no effect immediately on impact. We see this

as natural given the conflicting forces (inflation versus contraction in output) acting to raise and lower equity prices at the same time. The GBP devalues gradually against a basket of trading-partner currencies, reaching a trough of about 1 percentage point devaluation at longer horizons. The picture for both market variables is clearer in our monthly LPs.

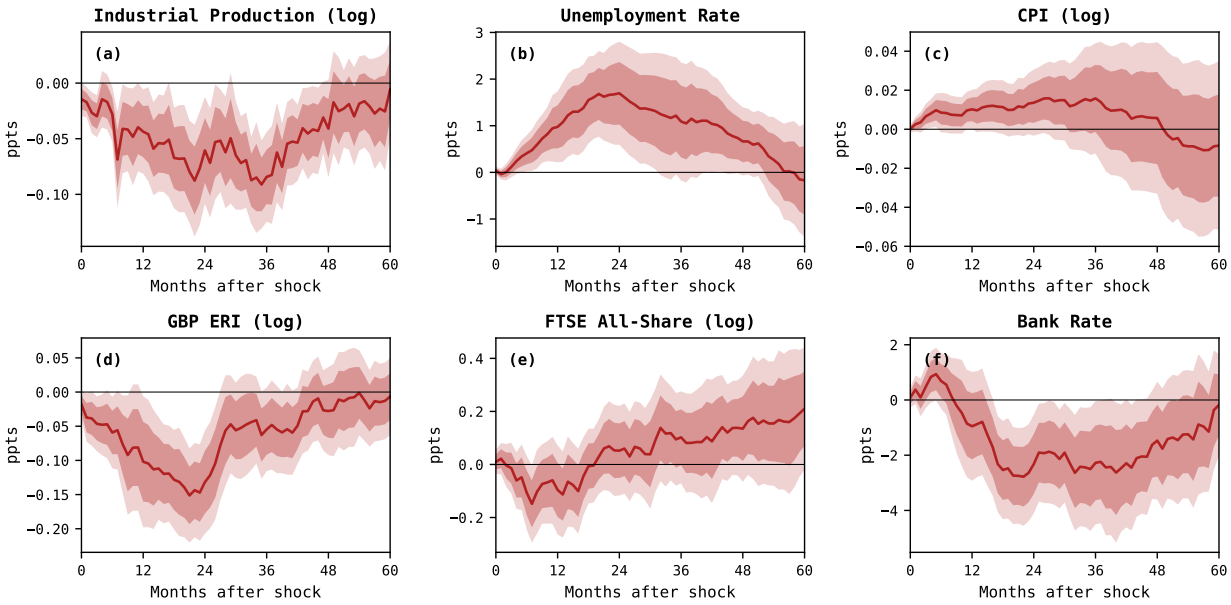
We use breakeven inflation as our high-frequency measure of market-based expected inflation. The breakeven is calculated using the difference between nominal and inflation-protected yields. The breakeven can be seen as a proxy for expected inflation over the following period equal in length to the maturity of the bond. For example the 3Y breakeven is a proxy for average inflation over the next 3 years.

We present responses of the 3, 5, and 10Y breakeven inflation rates. There is no shorter horizon possible given Breakeven construction. All three durations show persistent increases in breakeven inflation following the fiscal surprise. A fiscal surprise of 1 standard deviation pushes up yields by around 20-30 bpts (0.2-0.3 percentage points), and a corresponding breakeven inflation reaction of a similar magnitude if slightly smaller at peak.

6.3 Monthly Responses

Figure 5 presents impulse responses of monthly macroeconomic variables to a 25 basis point fiscal news shock. The results paint a coherent picture of contractionary transmission operating through financial conditions, with a secondary inflationary impulse driven by the exchange rate rather than domestic demand.

Industrial production falls sharply on impact, reaching a trough of approximately 10 percentage points below baseline around months 12–18, before recovering gradually over the five-year horizon. The response is precisely estimated throughout, with the 68 percent confidence band remaining below zero for the first three years. The unemployment rate rises significantly, peaking at around 1 to 1.5 percentage points above baseline at roughly 18 months, and remaining elevated



Note: Dynamic responses estimated according to the equation (9) above. Shaded areas represent 68 and 90 percent confidence intervals respectively. Standard errors are autocorrelation-robust up to $h + 1$ lags.

Figure 5: Responses of Monthly Macro Variables to a Fiscal News Shock

well into the third year. Together, these labour market and output responses confirm the contractionary demand-side reading of the fiscal news shock.

The aggregate CPI response rises by approximately 1 percentage point on impact, reflecting the pass-through of Sterling depreciation into import prices in the months immediately following the shock. This initial inflationary impulse fades over the medium term as the demand contraction increasingly dominates, pulling the aggregate index back toward baseline by the second and third years. This speed of the initial inflationary response is informative about the transmission channel. Monetary policy transmits to prices slowly, operating through domestic demand, labour market tightening, and wage-setting cycles — a chain that typically takes 18 to 24 months to materialise fully in standard estimates (Christiano et al. 2005). Fiscal news, by contrast, transmits to prices rapidly through the exchange rate: Sterling depreciation raises the domestic currency cost of imported goods almost immediately, and retailers pass on these cost increases within one to two

quarters as import prices at the dock adjust (Burstein and Gopinath 2014; Gopinath et al. 2010).

As Figure 7 shows, the aggregate masks sharply divergent dynamics across the consumption basket: goods prices rise significantly and persistently, peaking at around 1 to 1.5 percentage points on a year-on-year basis by month 24, while services prices are essentially unchanged throughout the horizon. Since services constitute the majority of the CPI basket and are driven primarily by domestic wage costs rather than import prices, the aggregate index is pulled toward zero at medium horizons even as tradeable goods inflation remains elevated. This decomposition is central to identifying the exchange rate pass-through channel and is discussed further in Section 5.5.

Sterling depreciates significantly, with the GBP ERI falling by around 15 percentage points at the trough, consistent with a "risk-off" reassessment of UK sovereign creditworthiness rather than a Mundell-Fleming appreciation in response to higher yields.

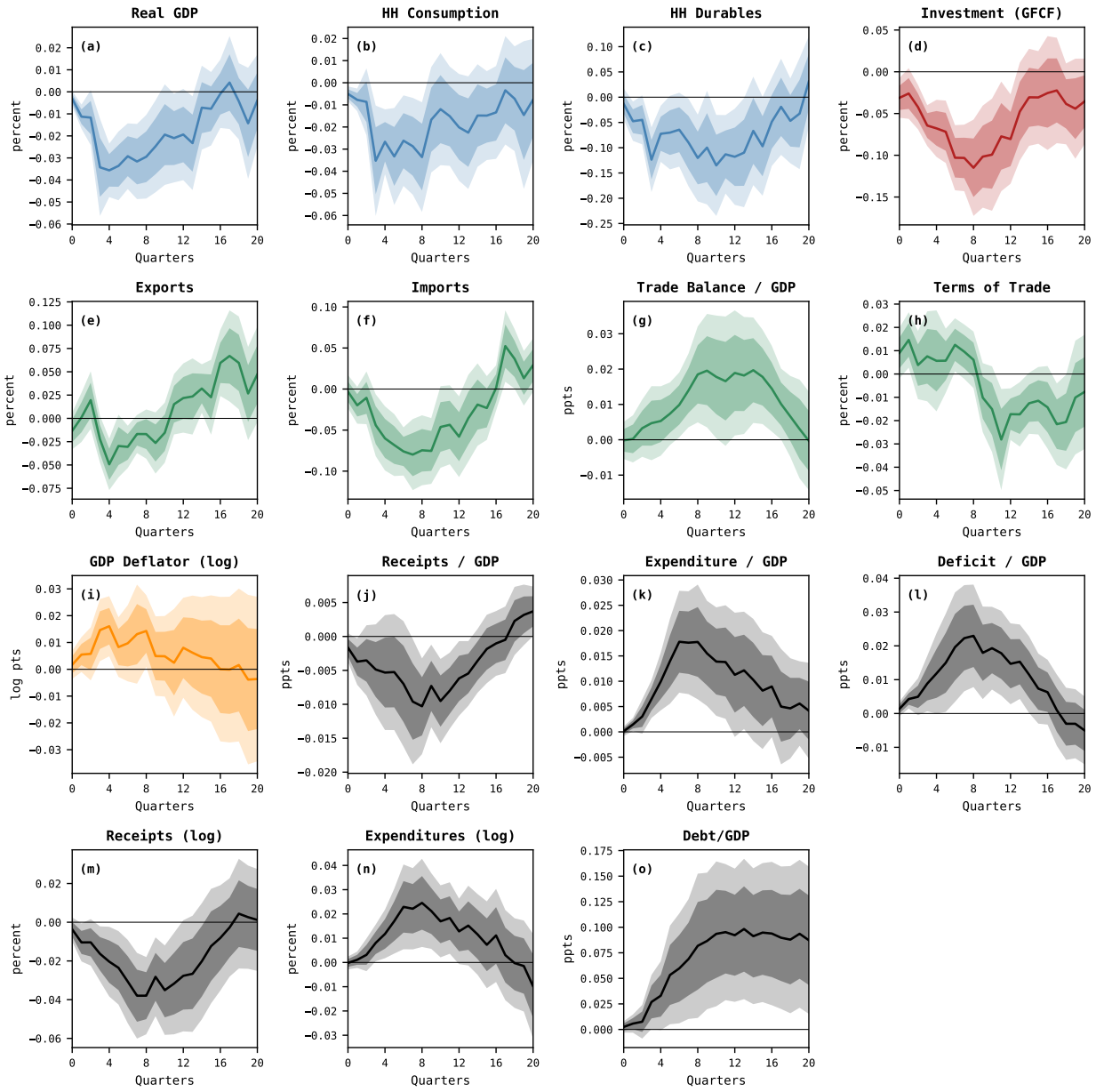
The Bank Rate responds by tightening in response to CPI pressure in the first year or so after the fiscal news shock, before turning negative at medium horizons, reflecting the BoE's stagflationary trade-off - facing rising unemployment and rising prices simultaneously.

6.4 Quarterly Responses

Figure 6 presents quarterly impulse responses across fifteen outcome variables organised by demand component, trade flows, prices, and fiscal accounts. The quarterly results corroborate the monthly evidence while providing additional detail on the composition of the demand contraction and the fiscal consequences of the shock.

Real GDP falls persistently, reaching a trough of approximately 3 percentage points below baseline around quarters 4 to 8, before showing signs of gradual recovery. The contraction is broad-based across demand components.

Investment (GFCF) exhibits the largest and most precisely estimated response, falling by around 10 percentage points at peak, consistent with a sharp rise in the user cost of capital as gilt yields



Note: Dynamic responses estimated according to the equation (9) above. Shaded areas represent 68 and 90 percent confidence intervals respectively. Standard errors are autocorrelation-robust up to $h + 1$ lags.

Figure 6: Quarterly Responses to a Fiscal News Shock

rise, credit conditions tighten, and imported inputs rise in price. Household consumption declines by a similar order of magnitude to GDP, while durable expenditure falls by more than non-durable consumption, as expected given the interest-sensitivity and lumpiness of large discretionary purchases. Households delay large discretionary purchases, consistent with weak economic activity and tighter credit conditions depressing durable expenditure disproportionately.

Trade dynamics are more complex. Exports display typical J-curve effects - exports contract as short-run financial conditions bite in the first year or so after the shock. Exports recover and turn positive by around the third year after impact as the competitiveness gain from sterling depreciation works through to export volumes. This delayed export response is consistent with standard J-curve dynamics in which contracts are pre-set in foreign currency and quantity adjustment lags price adjustment by several quarters.

Imports decline on impact and reach a trough at around the end of the second year as the combined effect of compressed domestic demand and higher sterling import prices reduces import volumes, before recovering gradually toward baseline as the demand contraction unwinds.

The terms of trade, measured as the ratio of the export to import price deflator, exhibit a two-phase dynamic. In the first phase, Sterling depreciation raises export prices in domestic terms faster than import prices adjust. After the end of the second year after the fiscal sustainability shock this balance reverses and the terms of trade falls up to 2 percentage points below initial levels, before slowly recovering by the end of the horizon.

Consistent with monthly CPI inflation-then-reversal dynamics, the GDP deflator rises to almost 2 percentage points above baseline around the start of the second year after the shock, before a strong reversal as the worsening macro environment depresses demand.

Government expenditure rises as a share of GDP, driven by automatic stabilisers, while receipts fall as output and employment contract. The primary deficit widens, reaching around 2 percentage points of GDP by the end of the second year. The debt-to-GDP ratio accumulates broadly in line

with the implied deficit flows, rising by approximately 10 percentage points cumulatively over five years. This ex-post fiscal deterioration validates the market's initial reassessment at the time of the shock, and is consistent with the self-reinforcing feedback between fiscal expectations, sovereign risk premia, and macroeconomic outcomes discussed in Section 5.6.

6.5 Imported Inflation: Goods versus Services CPI

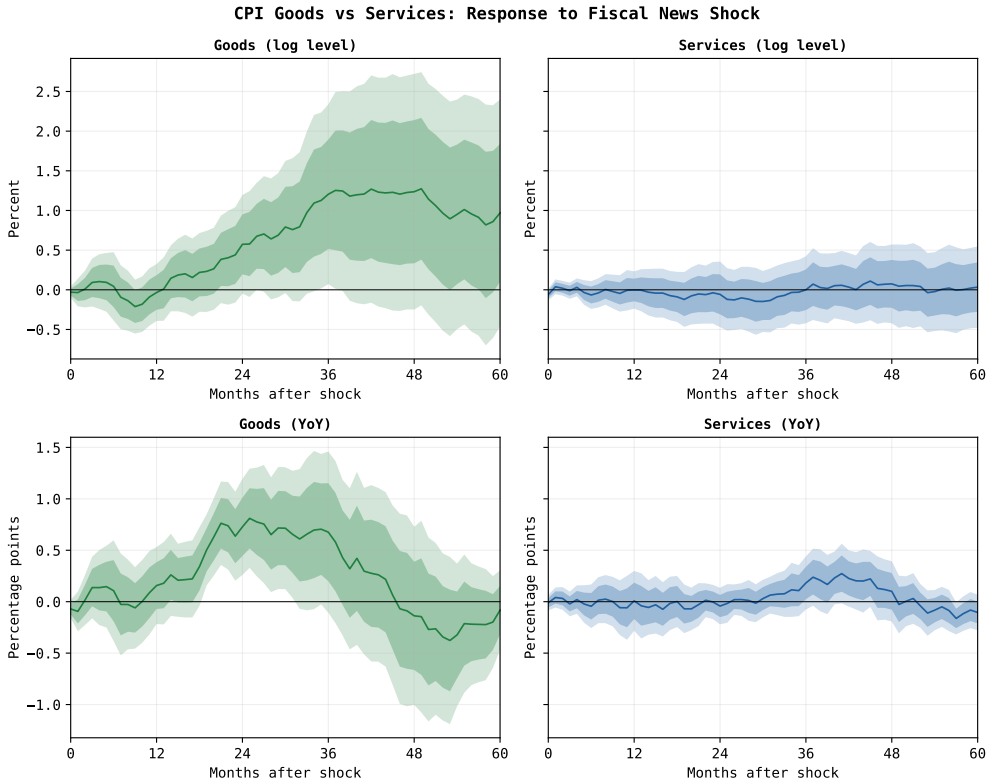
We see the channels through which the fiscal news shock transmits to the real economy clearly through consumer prices. Following the depreciation of sterling and the tightening of financial conditions, the transmission to consumer prices operates primarily through the traded goods component of the CPI basket. The fiscal news shock has effects confined almost entirely to the goods portion of the CPI basket, with Services CPI showing almost no response in comparison.

The UK is a small, relatively open economy. These responses are consistent with a stylised model in which goods are imported and their local price depends on exchange rate movements, while services are domestically produced and not tradeable, with a price largely set by domestic wage pressures. Firms' marginal cost depends on world price of inputs p_t, the exchange rate e_t, domestic wages w_t and crucially the import intensity of production α_M. UK goods feature a high import intensity.

6.6 Discussion

The pattern of impulse responses points to a transmission mechanism that operates through several reinforcing channels, and which differs in important respects from standard monetary policy transmission. The primary demand channel is conventional: tighter fiscal expectations raise borrowing costs, depressing business investment sharply and persistently, with household consumption and durable expenditure following with a lag.

What is less conventional is the behaviour of the exchange rate and inflation expectations.



Note: Dynamic responses estimated according to the equation (8) above. Shaded areas represent 68 and 90 percent confidence intervals respectively. Standard errors are autocorrelation-robust up to $h + 1$ lags. The consumer price index and its components is available only from 1988

Figure 7: Response of Prices and Inflation to a Fiscal News Shock

Rather than appreciating in response to higher gilt yields (as Mundell-Fleming logic would predict) Sterling depreciates on impact. We interpret this as evidence that markets read the fiscal surprise as a signal of deteriorating sovereign creditworthiness. The depreciation feeds directly into import prices, which rise persistently (as evidenced by CPI in goods alone rising, not in services), and generates a cost-push inflationary impulse that reinforces the demand contraction.

Most strikingly, breakeven inflation rates rise at all horizons – one, five, and ten years – following the fiscal shock. A purely cyclical or transitory explanation (exchange rate effects alone) would predict a front-loaded, short-end response as import cost pass-through works through the price level. The broad-based and persistent rise across the term structure of the breakeven inflation curve is instead consistent with a fair degree of markets re-pricing fiscal risk into long-run inflation expectations, reflecting concerns about the sustainability of the public finances and the possibility of partial future monetisation to maintain the government's budget constraint. This pattern is reminiscent of the fiscal dominance channel formalised by Sargent and Wallace 1984 and Cochrane 2023, and echoes the empirical evidence on fiscal risk premia in sovereign bond markets Laubach 2009; Gruber and Kamin 2012.

The fiscal accounts themselves reinforce this reading. The shock generates a deterioration in the primary deficit as automatic stabilisers activate, with debt accumulating broadly in line with the implied deficit flows. The market's initial reassessment of fiscal risk is therefore validated ex post by the realised fiscal dynamics, suggesting a potential degree of self-fulfilling feedback between fiscal expectations, risk, and macroeconomic outcomes that standard demand-side frameworks do not capture.

The estimated responses are quantitatively meaningful: at the baseline normalisation of a 25 basis point surprise (with Bank Rate endogeneously responding up to 1 percentage point), peak GDP and consumption fall by approximately 3 percent, while investment and consumer durables fall by close to 10 percent, magnitudes broadly comparable to estimates of monetary policy trans-

mission over similar horizons. These findings are robust across a range of specification checks, to which we now turn.

7. Robustness

We conduct several robustness checks to assess the reliability of our baseline findings. Across alternative estimation windows, shock definitions, treatments of outlier observations, and identification schemes, the baseline impulse responses are broadly stable.

Predictability of Fiscal News Shocks We confirm that the fiscal news shocks are unpredictable conditional on the covariates in our preferred specification. At the daily frequency, the yield surprise on fiscal event days cannot be predicted using information in the five trading days preceding the announcement. At the quarterly frequency, we residualise the fiscal surprises on own lags, together with one year of lags of output growth, inflation, unemployment, and the short-term interest rate. We then test the conditional exogeneity of the resulting shock by regressing it on twelve lags of each predictor separately. Only one predictor out of twelve is significant at the ten percent level — fewer than the number expected by chance under the null of no predictability. The distribution of p-values is approximately uniform on $[0, 1]$, consistent with $Pr(P < p) = p \in [0, 1]$. We repeat this exercise conditioning on event-quarter observations only, though with approximately one third of the observations this test has substantially lower power.

Alternative Estimation Periods Our baseline results are robust to extending the sample to include the COVID-19 period and the Truss–Kwarteng Mini-Budget shock of September 2022, from 1980 to 2025 inclusive. A practical limitation is that each event contributes a complete event-IRF only if H quarters of outcome data are available following the shock date. For the 2022 shock this requires national accounts data through end-2026, which at present is not available. This

ragged-edge problem is well documented in the local projections literature (see e.g.Jordà 2005). Differences in the IRFs in activity (GDP, consumption, investment) exhibit the classic midway cutoff due to incomplete horizon. As a result, the full OLS extended sample places extreme weight on the post-2022 macro dynamics at intermediate horizons, since the September 2022 shock is an approximately 6 standard deviation event relative to the distribution of Budget-day surprises in the baseline sample (see Table 3) and therefore dominates the OLS weighted average of event-IRFs. When we instead apply a Gaussian kernel that smoothly down-weights shock realisations in proportion to their size, but most strongly for very large shocks of either sign, assigning a weight of approximately 0.51 to the Truss–Kwarteng event, the full-sample IRFs converge closely to the baseline. We interpret this as confirmation that the baseline results are not an artefact of sample selection, and that the unusual dynamics of the 2022 episode reflect the extreme and atypical nature of that shock rather than a structural shift in fiscal transmission.

Our results are also robust to excluding the period of formal or informal fixed exchange rate with major European trading partners (1987-1988, 1990-1992). IRFs do not substantially change, and so we conclude transmission to the real economy is not effectively altered by a floating or (attempted) fixed exchange rate regime.

Alternative Treatments of Outlier Shocks The local projection estimator admits a natural representation as a weighted aggregation of individual event dynamics. Formally, the impulse response at horizon h satisfies

$$\hat{\beta}^h = [constant] \cdot \sum_{t \in \mathcal{E} \setminus \mathcal{X}; (t+h) \notin \mathcal{X}} s_t \cdot \beta_t^h = \mathcal{A}(s_t, y_t, h, \mathcal{E}, \mathcal{X}) \tag{10}$$

where $\mathcal{E}$ denotes the set of Fiscal event dates and $\mathcal{X} := \{t : t \in \text{excluded dates}\}$ is a set of dates excluded from both the left- and right-hand sides of the regression. Observations in $\mathcal{X}$ contribute neither as regressors nor as outcome long-differences. We exclude two windows: the COVID-19

period (2020Q1–2021Q4) and the Mini-Budget episode (2022Q3–2022Q4), the latter defined to include both the Kwarteng announcement. The IRFs are qualitatively unchanged relative to the baseline, and notably smooths over the ragged-edge problem without sacrificing observations.

Rather than exclude outlier observations entirely, we also estimate the local projections with a soft down-weighting kernel that assigns weights close to one for most shocks, but progressively discounts extreme realisations.

Alternative Maturity of Surprises Our results are robust to using the 1Y gilt yield surprise or the surprise based on the first principal component across the (1,3,5,10) year yields in place of the three-year. The one-year yield is more closely anchored to Bank Rate expectations and may therefore capture a narrower component of fiscal news than the three-year, which better reflects the horizon of a parliamentary term. Quantitatively, the IRFs are close to baseline and our substantive interpretations are unchanged.

Alternative Shock-filtering Process We also employ the two-stage autoregressive filter of Miranda-Agrippino and Ricco 2021 in their informationally-robust shock construction. We regress the event-day fiscal surprise on one lag in event-time, to remove autocorrelation in market surprises. This first stage in the monetary literature captures FOMC information effects, while in fiscal policy we would argue that this is prudent behaviour on the part of the fiscal policymaker trying to smooth market disruptions. The second stage takes these event-AR1 residuals, and in calendar-time runs a monthly AR12 to remove any residual autocorrelation due to slow updating, and deeper slower-moving dynamics. The residuals from the second stage are then the shocks of interest. We would argue our one-stage process is the FWL equivalent with one notable difference - we control for more lags in events to more flexibly capture fiscal surprise autocorrelation. In a heuristic sense our baseline is the AR(p_e, p_m) equivalent to the MAR-type filter but with extended event-lags. While appropriate for their monetary setting, our fiscal surprises show a higher degree of persistence in

the earlier years of the sample, suggesting more than one event-lag is needed to fit true shocks. Indeed shocks fitted as described above with only one event-sequence-lag exhibit autocorrelations at the short end, while our baseline fitted shocks do not.

Ruling out Growth News in Official Forecast Revisions A concern with using gilt surprises is that they may reflect market reactions to (1) the government's superior information set revealed in the fiscal event, essentially signalling future macro conditions (2) the Office for Budget Responsibility (OBR)'s revised outlook for the UK economy captured by revisions to the short-to-medium run growth rate.

$$s_e = \alpha + \beta(\mathbb{E}_e[GDP^{OBR}] - \mathbb{E}_{e-1}[GDP^{OBR}]) + \epsilon_e \tag{11}$$

We show that gilt surprises on Budget days are not driven by markets learning about the GDP growth outlook from the OBR's forecasts. We do this by regressing the 3-year gilt surprise on the OBR's concurrent GDP growth forecast revisions at horizons of zero, one, and two years ahead, using all fiscal events accompanied by a contemporaneous OBR *Economic and Fiscal Outlook* over the period 2010–2025.[2] Figure 17 presents the results. The relationship is statistically insignificant at all horizons (adjusted $R^2 = 0.016$, $F = 1.141$, $p = 0.352$), and the point estimates are economically small: taking the regression coefficients at face value, and overlooking the problem these tests are likely underpowered, the largest OBR growth revision in our sample implies a fitted gilt surprise of at most 3.5 basis points. Moreover, even if the relationship were significant, the sign is negative, meaning negative growth news would raise yields, thus aligns with our interpretation of event-day surprises as sustainability news.

2. We exclude events without a contemporaneous OBR forecast, including the September 2022 Truss–Kwarteng mini-budget, spending reviews, and emergency fiscal statements, leaving $n = 29$ observations.

Ruling out Growth News with "Poor Man's Sign Restrictions" on Event Days The above exercise deals with growth news through updates to official growth forecasts. Next, we address the possibility government announcements contain growth news due to a superior government information set, and event day surprises are potentially confounding true fiscal news (based on sustainability concerns) with broader macro news updates encoded in the announcements.

We employ the fiscal event-day sign restriction identification of Jarociński and Karadi 2020: this decomposes fiscal surprises into two components depending on the comovement of yields and financial instruments: fiscal sustainability news (the analogue of monetary policy shocks) and fiscal growth news (the equivalent of a Fed information effect). The two are distinguished by whether yields and financial instruments move in the same or opposite directions at the time of the policy event.

In this scheme, a debt sustainability news shock is when yields and financial markets move in opposite directions, for example, yields rise following a negative sustainability news shocks, and the bilateral GBP-EUR exchange rate falls. The key here is that movements in the yield and the other financial indicator go in opposite directions, while aligned directions would represent something theoretically akin news about future growth. Expansionary fiscal growth news raises growth expectations, attracting capital inflows alongside higher real rates. This fiscal information or signalling channel is formalised in Melosi et al. 2024, who show that fiscal announcements convey government information about future economic conditions beyond the direct policy change, analogous to the central bank information shock of Jarociński and Karadi 2020.

In practice, one faces a large menu of financial instruments, all of them noisy, from which to choose. We use a selection of financial instruments (bilateral FX rates against USD, EUR (DEM before 1999 at the Euro peg), the GBP NEER index, and the FTAS stock index). One draw-back of PMSR in this context is that for any given instrument the fiscal news may simply be drowned out by secular movements in the financial instrument, essentially making the on-the-

day changes noisy representations of the true shocks. The ensemble of PMSR IRFs is broadly consistent with our baseline, noting that this approach typically halves our event day observations, therefore estimated dynamics have wide confidence intervals, and may be subject to swings due to dominating outside factors. Of all the PMSR IRFs, the EUR-identified series is notable as tracking our baseline extremely closely. The euro area accounts for around 40 percent of UK trade, therefore it can be expected that the GBP-EUR rate would be most responsive to UK-specific fiscal news. Overall, this PMSR exercise supports our baseline specification decisions, however, specifically in the case of Debt-to-GDP, our baseline response appears to be at the very upper end of what PMSR would otherwise suggest, and therefore some caution is needed when interpreting the Debt-GDP ratio IRF in isolation. Given fewer event days, wider confidence bands, and potential for sensitivity of the composite IRF to certain episodes, it should also be noted that PMSR may be under-powered. For brevity we do not report the precision-weighted model average, or the Bayesian hierarchical aggregation, but both conform to our baseline specification.

Together, these two exercises address the concern that gilt surprises on Budget days capture macroeconomic information rather than fiscal news. The OBR revision regression rules out that markets are repricing off the OBR's publicly released GDP growth outlook: the relationship between our yield surprise and the OBR's simultaneous GDP forecast revision is statistically insignificant and economically negligible. The Poor Man's Sign Restrictions address the complementary concern that the government itself conveys superior information about future growth conditions through the announcement: by restricting to Budget days where gilt yields, sterling, and equity prices comove in directions consistent with a fiscal sustainability shock — and against the pattern expected from growth news — we isolate events where a macroeconomic information interpretation is implausible. The two checks are complementary in scope. The OBR exercise rules out publicly revealed forecast information, while the PMSR rules out privately held government information revealed through the announcement itself. That both exercises yield results broadly

consistent with the baseline supports our interpretation that the gilt surprise primarily captures market reactions to the fiscal stance, debt sustainability, and credibility of the public finances.

A natural question is why the fiscal information channel appears relatively muted in the UK setting, given evidence of such effects elsewhere. We argue that the institutional design of UK fiscal policy mitigates the scope for endogenous announcement timing through which fiscal announcements can convey macroeconomic information. Unlike emergency fiscal measures, where the very act of calling an unscheduled announcement may itself signal deteriorating economic conditions, UK Budgets follow a largely predictable calendar-driven schedule. This limits the extent to which the timing of a Budget can be read as a signal of the government's private assessment of economic prospects. The government's information advantage is further constrained by the presence of the OBR, whose independent macroeconomic and fiscal forecasts are published simultaneously with the Budget and serve as the separate basis for fiscal planning. In this institutional environment, the marginal information conveyed by the announcement itself about future growth conditions is limited, and the yield surprise more cleanly reflects market reassessment of fiscal sustainability and debt dynamics.

8. Conclusion

Fiscal news is an important driver of macroeconomic dynamics. We develop a high-frequency identification strategy to examine the impact of fiscal news on the UK economy. Using variation in government bond yields in a tight window around regular and comprehensive UK government fiscal announcements, we identify fiscal news shocks. For a small, open economy such as the United Kingdom, fiscal surprises have considerable power. Markets react to negative news by selling off British assets (stocks, government bonds, domestic currency). The combined effects are stagflationary - output and activity are depressed by tighter financial conditions, while devaluation

imports inflation. Our baseline results are robust to a battery of checks including alternative sample windows, alternative treatments of outlier shocks or macro dynamics, and alternative shock constructions. In separate work we examine heterogeneity in fiscal news shock transmission based on state of the business cycle, sign of the shock, and monetary policy offset.

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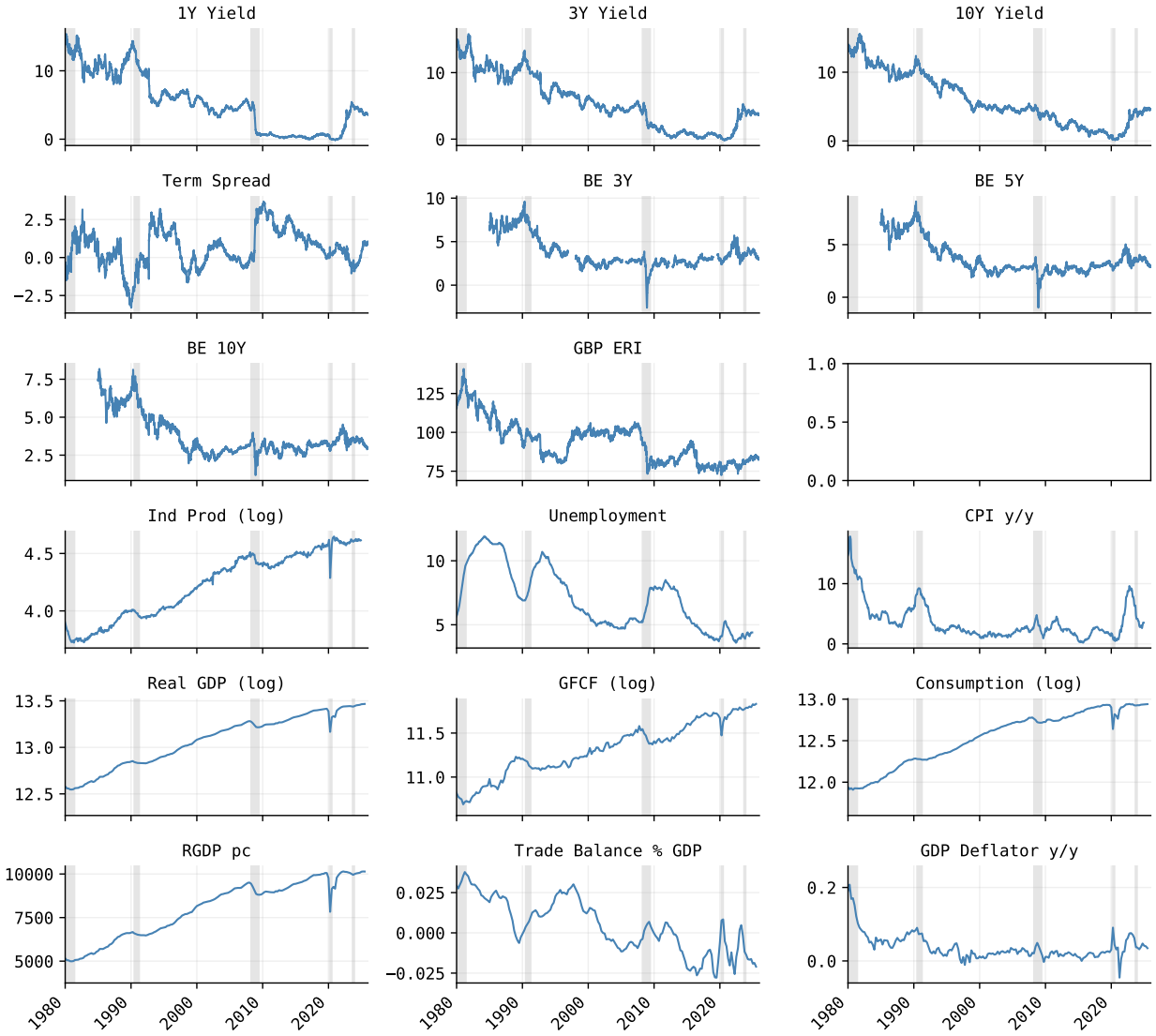
A. Additional Figures



Note: time series of main variables used in this analysis, mixed frequencies: daily, monthly, quarterly.

Figure 8: UK Macro Time Series

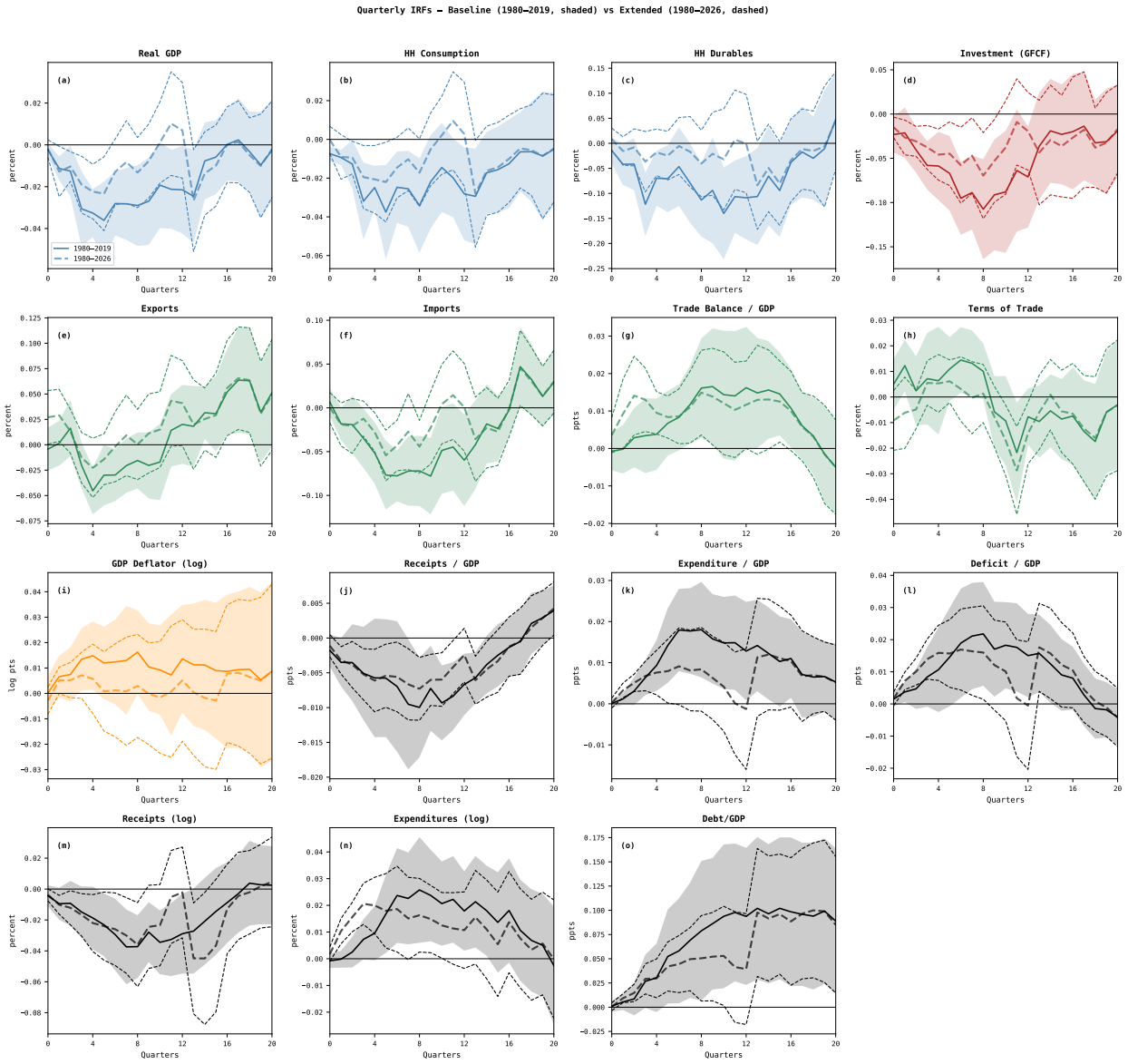


Figure 9: Alternative Estimation Windows: "Extended" Data Sample

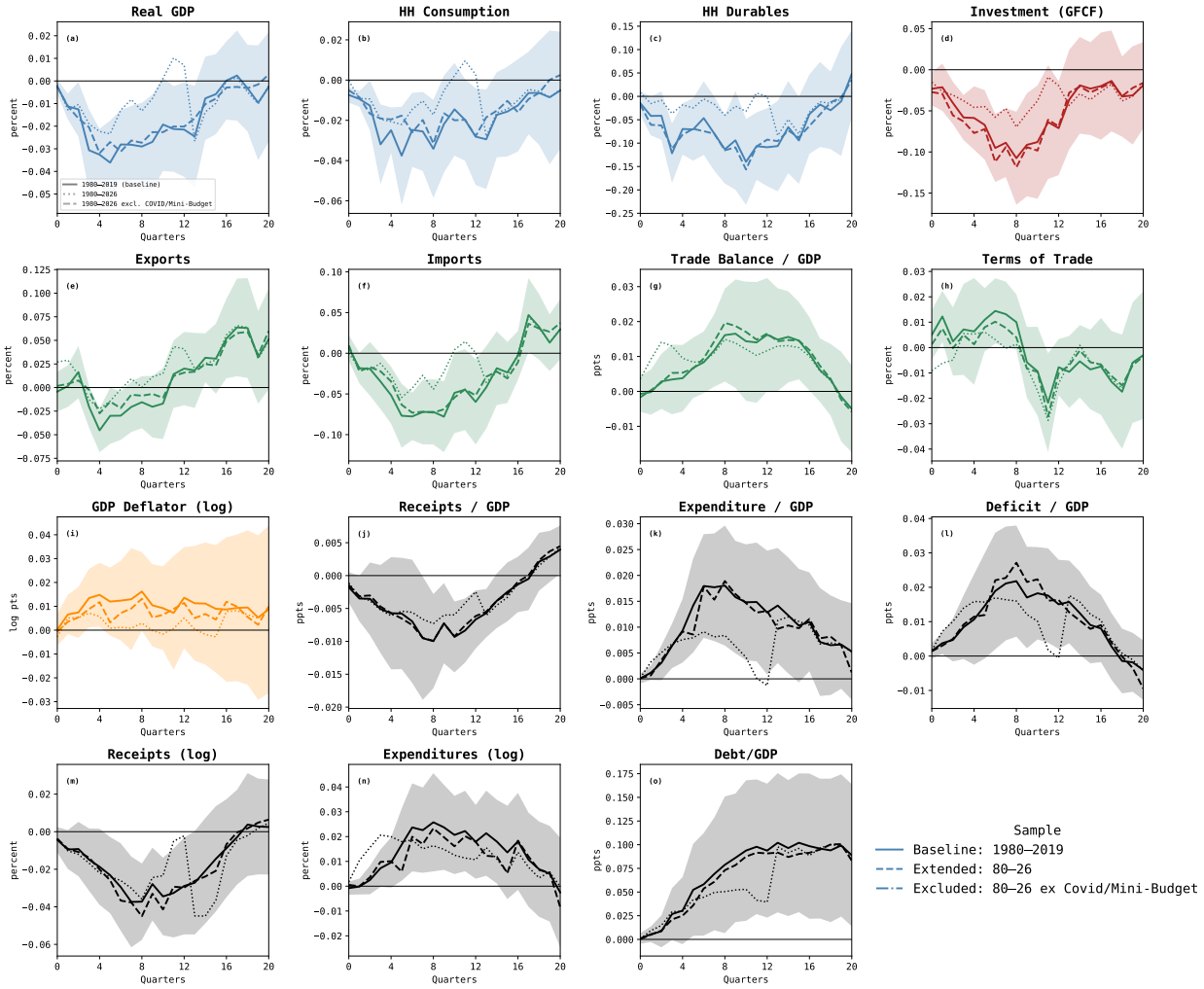
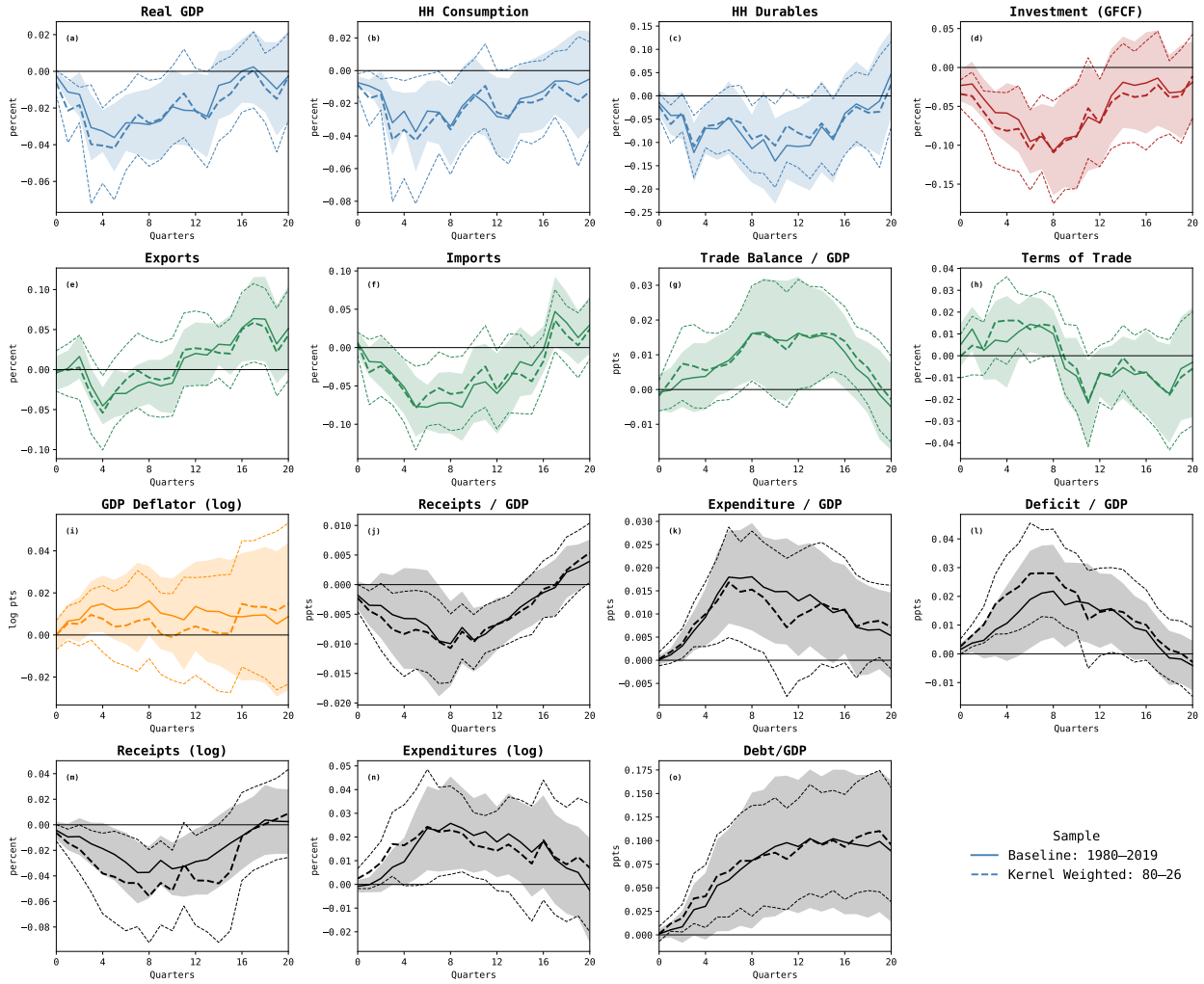


Figure 10: Alternative Estimation Windows: Baseline, Extended, Excluded



Note: In this robustness check we plot the baseline specification / sample in solid line/filled area against the alternative estimation using the full sample of data with no exclusions. We instead weight Fiscal surprises such that more extreme surprises have a reduced importance in the IRF estimation.

Figure 11: Alternative Estimation Weighting on Outliers: Baseline, Kernel Weighted

Let T denote the full sample size and define the following T-vectors of binary indicators:

$$e_t = \mathbf{1}\{t \in \mathcal{E}\} \quad x_t = \mathbf{1}\{t \in \mathcal{X}\} \quad x_t^h = \mathbf{1}\{t + h \in \mathcal{X}\} \tag{12}$$

with $\mathbf{e}, \mathbf{x}, \mathbf{x}^h \in \{0,1\}^T$. The horizon-$h$ selection matrix is:

$$\mathbf{P}^h \equiv \mathrm{diag}\big(\mathbf{d}^h\big)\,, \qquad \mathbf{d}^h = \mathbf{e} \odot (\mathbf{1} - \mathbf{x}) \odot (\mathbf{1} - \mathbf{x}^h) \tag{13}$$

so that $d_t^h = 1$ if and only if observation t is a valid event date and its horizon-h outcome does not fall within an excluded window. To downweight outlying shock realisations without discarding them, we augment $\mathbf{P}^h$ with a Gaussian kernel weighting matrix:

$$\mathbf{\Omega}^h \equiv \mathrm{diag}\big(\mathbf{w}^h\big)\,, \qquad w_t^h = d_t^h \cdot \exp\left(-\frac{s_t^2}{2c^2}\right) \tag{14}$$

where no observation is assigned zero weight by the kernel. The impulse response estimator at horizon h is then:

$$\hat{\beta}^h = \big(\mathbf{s}'\mathbf{\Omega}^h\mathbf{s}\big)^{-1}\,\mathbf{s}'\mathbf{\Omega}^h\Delta_h\boldsymbol{y} \tag{15}$$

which is the WLS coefficient from regressing $\boldsymbol{\beta}^h$ on $\mathbf{s}$ with weights $\mathbf{\Omega}^h$. The estimator nests the standard local projection as a special case when $c \to \infty$ (uniform weights) and the exclusion-only estimator when $c \to \infty$ with $\mathbf{\Omega}^h = \mathbf{P}^h$.

For example, the Truss-Kwarteng mini budget would receive a weight of $\exp(-0.49/2 \cdot 2 \cdot 0.09) = 0.51$ when the parameter of the weight is set to approximately 2 standard deviations of the surprise series on event days.

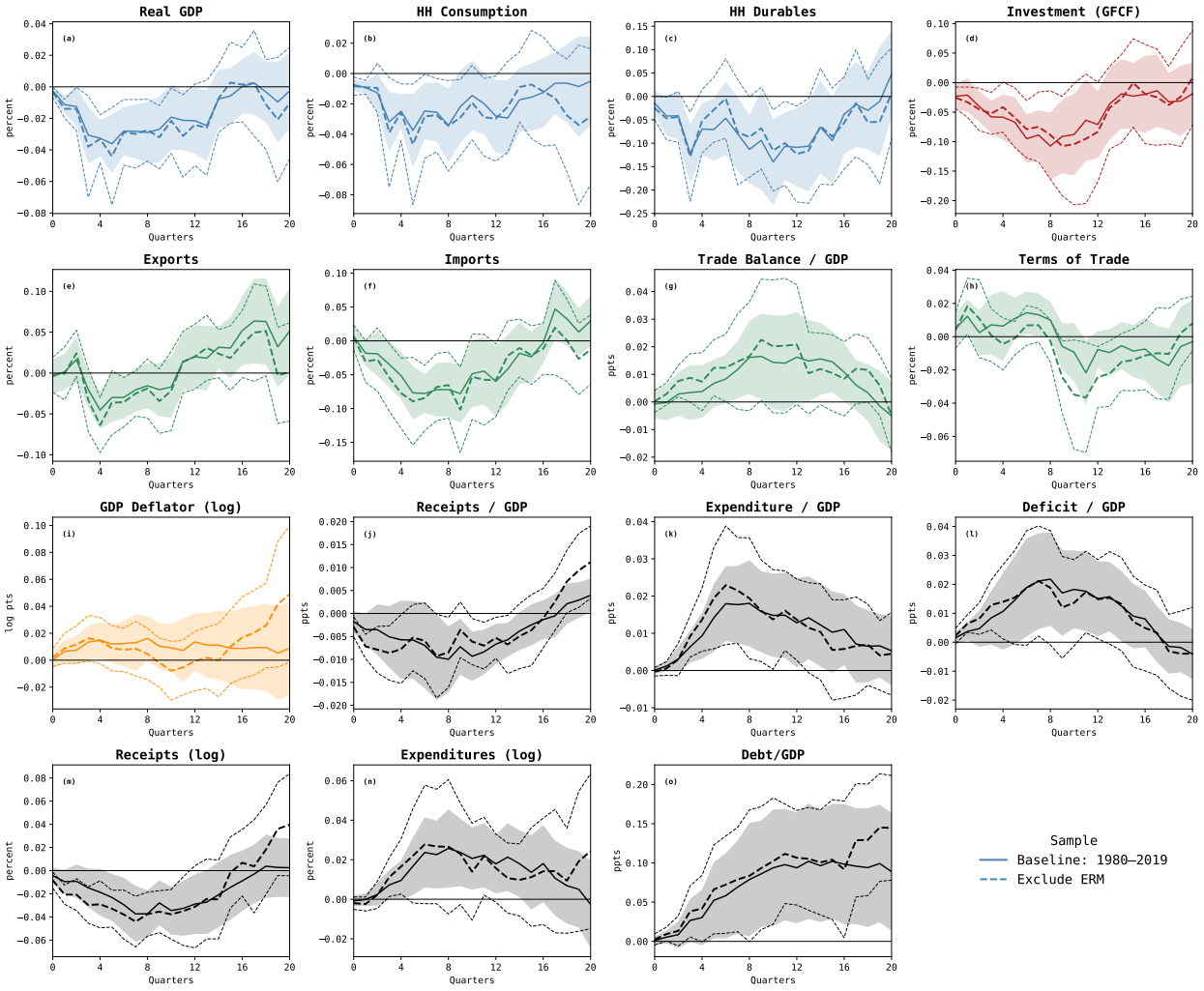


Figure 12: Alternative Estimation Windows: Baseline, Exluding ERM and informal peg period

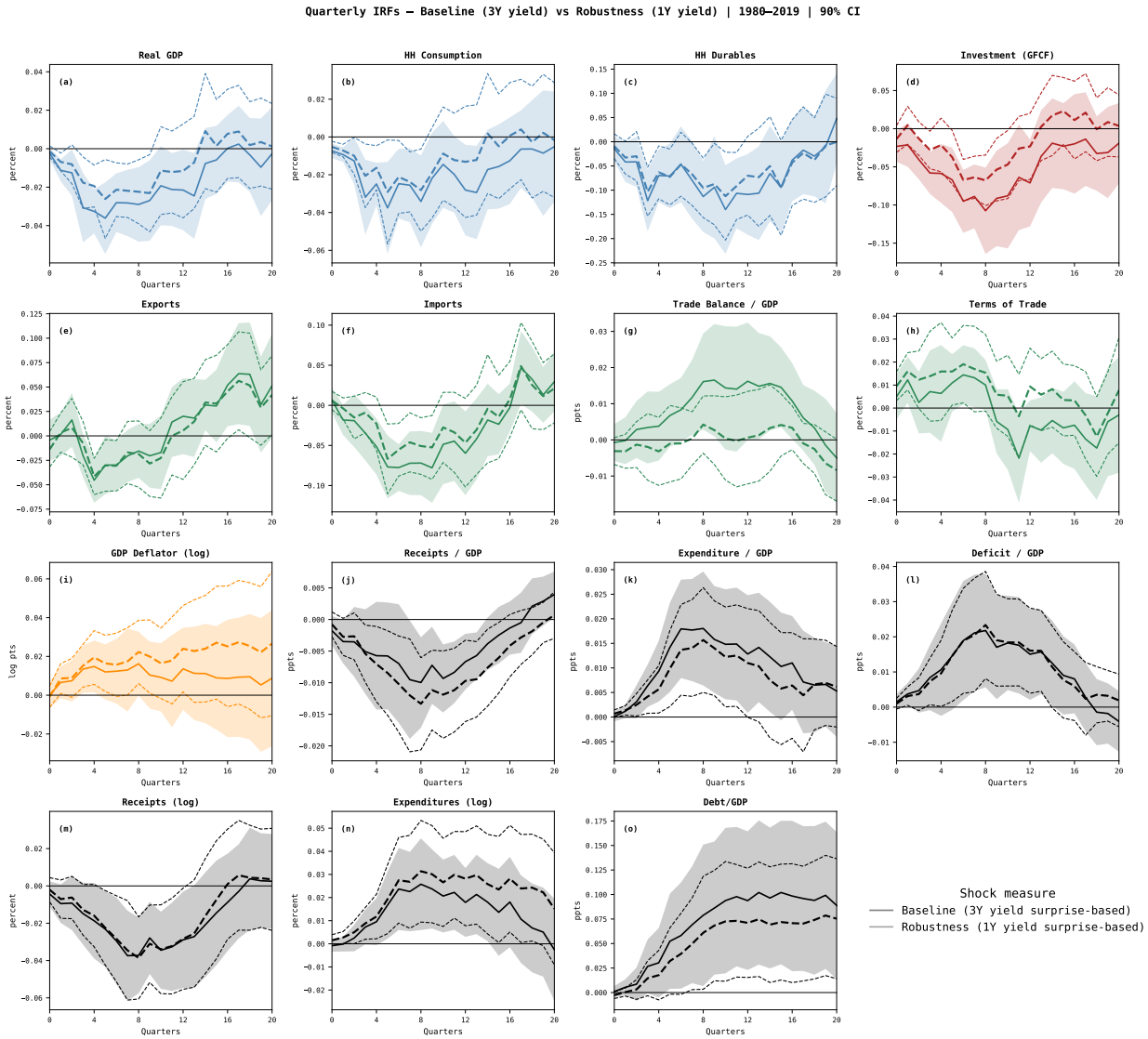


Figure 13: Alternative Shocks based on 1Y yield surprises

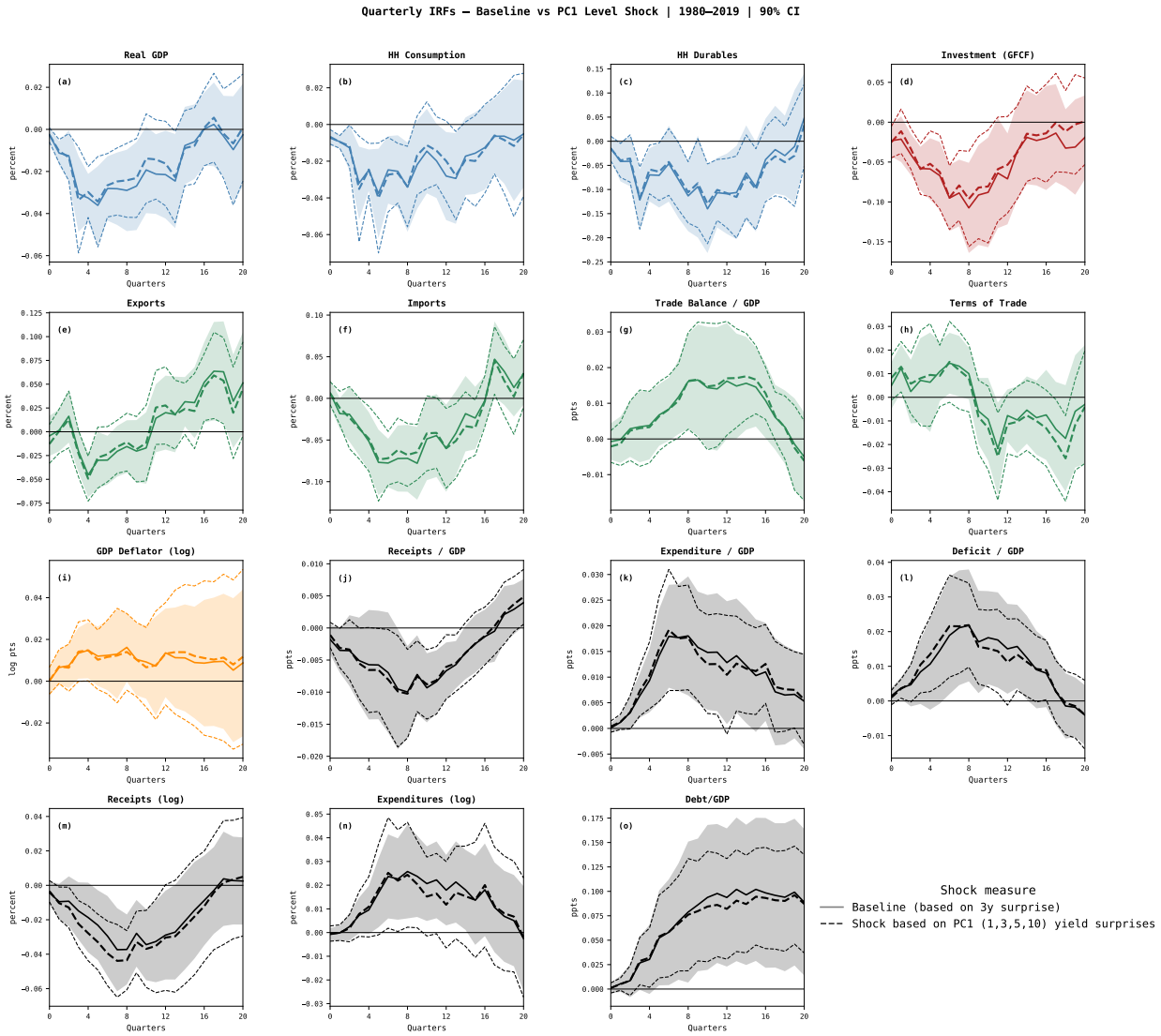


Figure 14: Alternative Shocks based on First Principal Component of (1,3,5,10) yields

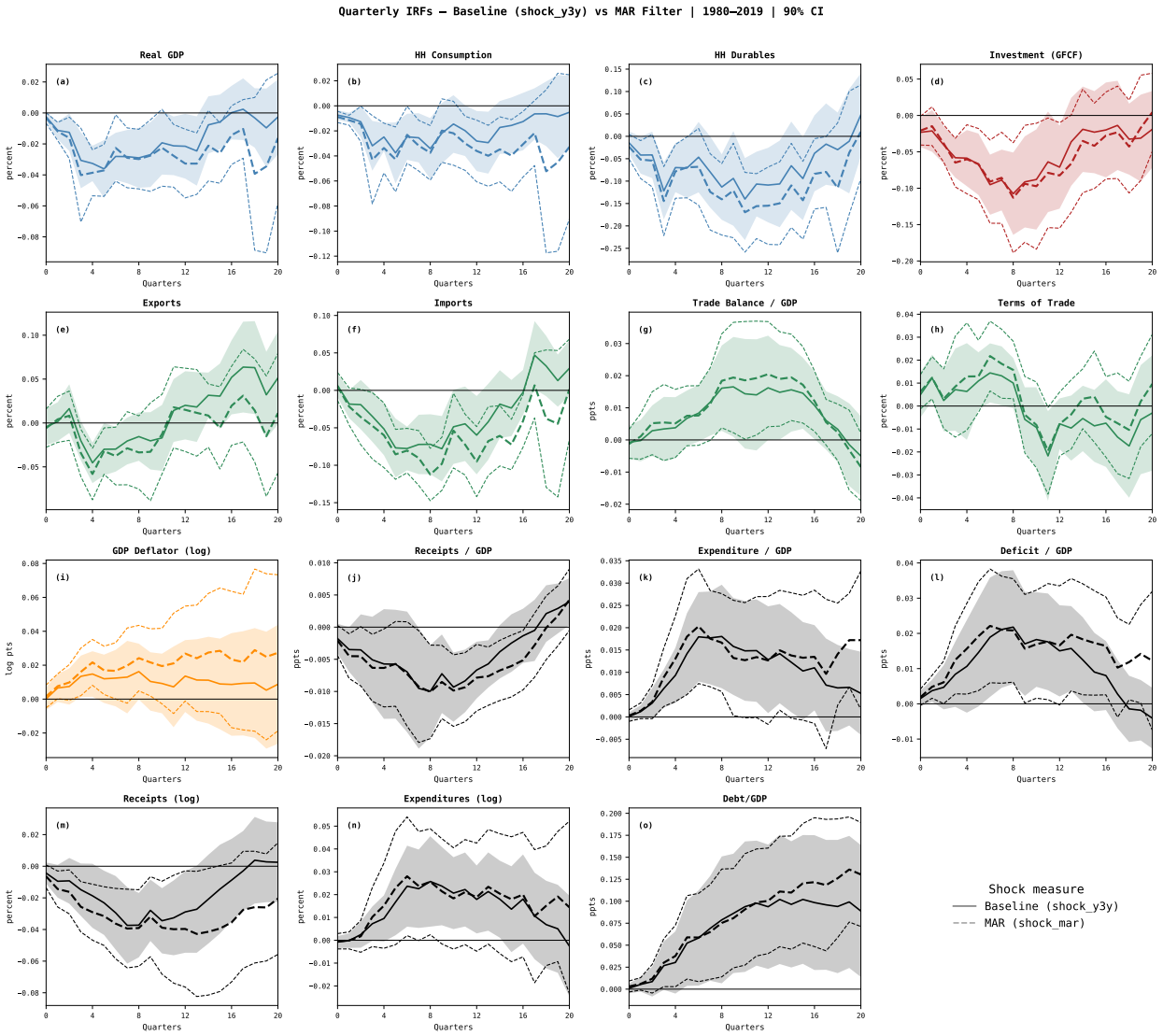


Figure 15: Alternative Shocks based on Miranda-Agrippino and Ricco 2021 AR($1_e, 12_m$) filter

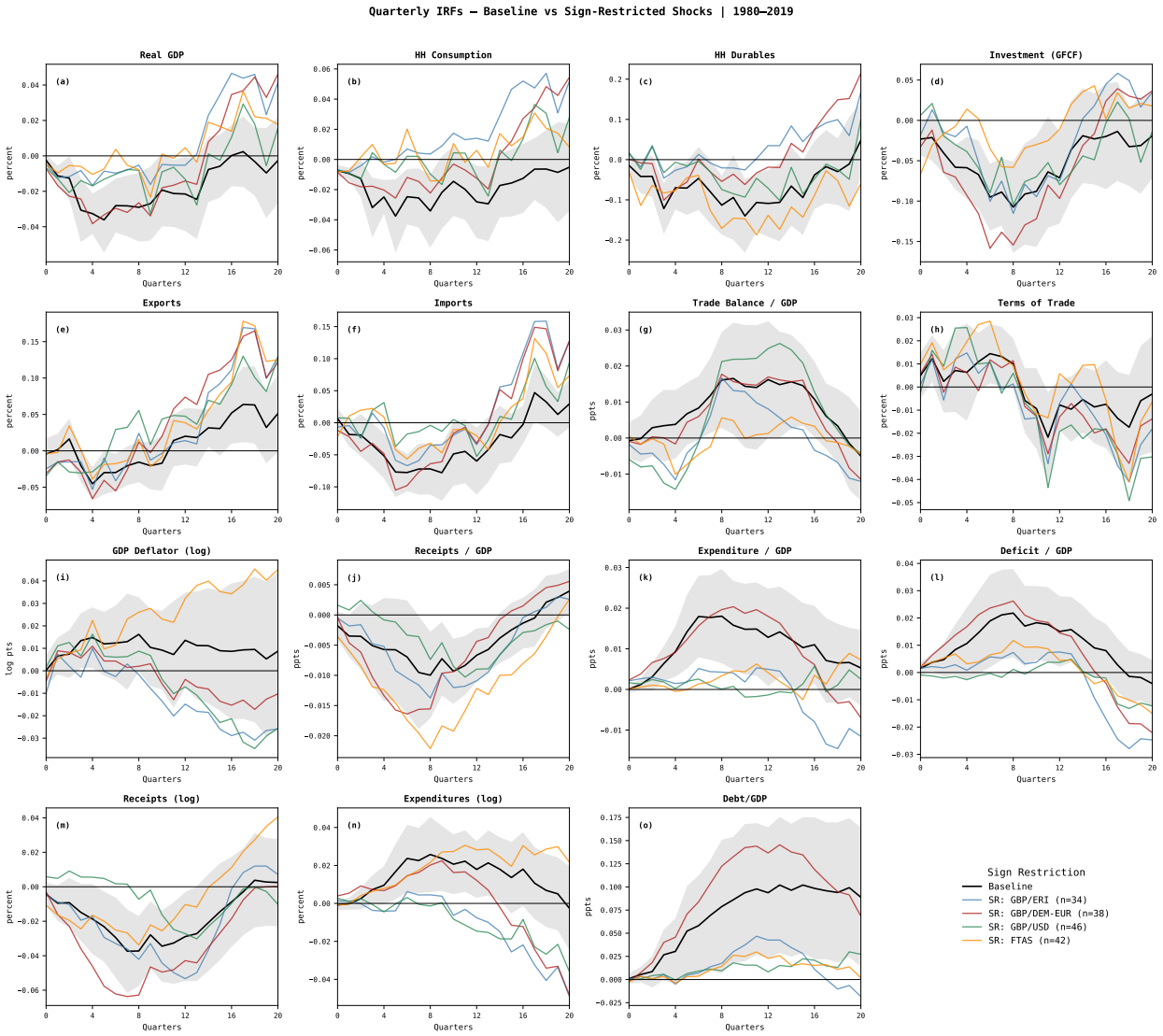


Figure 16: Alternative Shocks based on Jarociński and Karadi 2020 "Poor Man's Sign Restrictions"

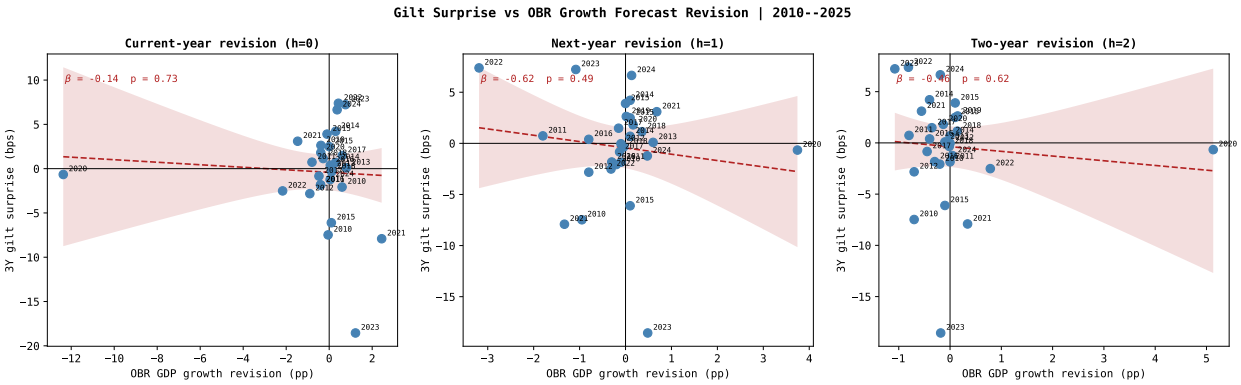


Figure 17: Scatter plots of OBR GDP revisions and Event Day Surprises

B. Additional Tables

Yield Maturity, m	1Y	3Y	5Y	10Y	PC1
Panel A: Baseline sample, 1980–2019					
N	72	72	72	72	72
Mean (bpts)	-0.82	-0.27	-0.12	0.31	-0.13
SD, event days (bpts)	6.15	7.06	6.68	5.80	1.75
SD, control days (bpts)	4.80	5.37	5.56	5.76	1.44
Ratio (event/control)	1.28	1.31	1.20	1.01	1.22
Min (bpts)	-20.21	-21.59	-21.75	-20.83	-5.95
date	Mar 1983	Mar 1983	Mar 1987	Mar 1987	Mar 1987
Max (bpts)	14.71	17.88	14.60	16.21	4.06
date	Mar 2008	Nov 2001	Nov 2001	Apr 2009	Nov 2001
AR(1)	0.31	0.18	0.16	0.06	0.20

Table 2: *Fiscal Surprise Summary Statistics.*

Yield Maturity, m	1Y	3Y	5Y	10Y	PC1
Panel B: Full sample, 1980–2025					
N	86	86	86	86	86
Mean (bpts)	-0.38	0.25	0.34	0.47	-0.01
SD, event days (bpts)	7.17	8.81	8.37	6.80	2.16
SD, control days (bpts)	4.88	5.46	5.63	5.86	1.47
Ratio (event/control)	1.47	1.61	1.49	1.16	1.47
Min (bpts)	-20.21	-21.59	-21.75	-20.83	-5.95
date	Mar 1983	Mar 1983	Mar 1987	Mar 1987	Mar 1987
Max (bpts)	33.21	49.64	46.91	29.10	11.51
date	Sep 2022	Sep 2022	Sep 2022	Sep 2022	Sep 2022
AR(1)	0.20	0.09	0.07	-0.01	0.09

Table 3: *Fiscal Surprise Summary Statistics.*

Horizon	0	20	40	60	80	100	Peak	h^*
Yields								
1Y Gilt Yield	6.87**	3.71	15.74**	21.22**	14.28**	13.96*	21.57**	59
	(0.40)	(7.69)	(5.46)	(7.14)	(8.36)	(11.49)	(6.99)	
3Y Gilt Yield	8.80**	8.49*	17.50**	27.04**	18.80**	16.63*	27.27**	61
	(0.11)	(5.84)	(5.53)	(8.37)	(9.28)	(14.01)	(8.27)	
5Y Gilt Yield	8.09**	8.81**	14.96**	24.93**	17.71**	13.87*	24.93**	60
	(0.35)	(4.88)	(5.38)	(7.56)	(8.28)	(12.81)	(7.56)	
10Y Gilt Yield	5.70**	2.70	6.19*	13.29**	6.07	0.62	13.29**	60
	(0.74)	(6.31)	(5.60)	(5.96)	(6.63)	(7.41)	(5.96)	
Exchange Rate & Equity								
GBP ERI (log)	-6.90	-33.03	-73.02**	-74.57*	-74.69*	-67.15*	-93.32**	57
	(7.11)	(39.13)	(41.05)	(49.38)	(52.91)	(56.92)	(54.84)	
FTSE All-Share (log)	26.62*	34.18	-7.49	-87.89*	-111.80*	-68.34	-155.61**	72
	(23.10)	(56.69)	(79.62)	(78.24)	(106.52)	(81.14)	(86.80)	
Breakeven Inflation								
3Y Breakeven	4.84**	11.84**	4.75	16.70**	12.61*	2.10	17.42**	61
	(1.53)	(5.78)	(6.30)	(7.35)	(9.79)	(12.79)	(7.45)	
5Y Breakeven	4.19**	12.10**	8.39*	16.80**	12.28*	2.29	16.95**	61
	(0.99)	(5.18)	(5.18)	(6.35)	(7.96)	(10.38)	(6.42)	
10Y Breakeven	3.12**	10.91**	9.74**	14.40**	9.46*	3.34	14.40**	60
	(0.93)	(3.89)	(4.07)	(5.58)	(6.54)	(6.94)	(5.58)	

Table 4: *Daily Impulse Responses to a 1 SD Fiscal News Surprise.*

	0	12	24	36	60	Peak	h^*
Industrial Production	-1.52*	-3.89*	-5.84**	-8.55**	-1.28	-9.31**	35
	(1.14)	(2.95)	(3.50)	(3.21)	(3.52)	(3.40)	
Unemployment Rate	0.05	0.95*	1.66**	1.29*	0.14	1.66**	24
	(0.06)	(0.63)	(0.90)	(1.06)	(1.10)	(0.90)	
CPI (log)	0.03	1.14*	1.39	1.20	-2.17	-2.42	57
	(0.13)	(0.80)	(1.71)	(2.86)	(4.53)	(4.48)	
Goods CPI (YoY)	-0.12*	0.07	0.70*	0.68*	0.08	0.76**	25
	(0.08)	(0.32)	(0.43)	(0.60)	(0.42)	(0.45)	
Services CPI (YoY)	-0.02	0.05	-0.01	0.12	-0.12	-0.24*	57
	(0.05)	(0.21)	(0.11)	(0.15)	(0.17)	(0.15)	
GBP ERI (log)	-1.95*	-8.33*	-10.77**	-1.84	1.65	-13.21**	21
	(1.51)	(6.92)	(6.45)	(5.03)	(6.12)	(5.76)	
FTSE All-Share (log)	0.41	-9.87	4.86	6.01	18.50**	18.50**	60
	(3.22)	(16.59)	(12.28)	(12.02)	(10.80)	(10.80)	
Bank Rate	0.04	-0.41	-1.38	-1.61*	0.54	-1.81*	32
	(0.27)	(1.60)	(1.53)	(1.44)	(1.79)	(1.47)	

Table 5: *Monthly Impulse Responses to a 25bp Fiscal News Shock.* Columns report point estimates at selected horizons (months) with standard errors in parentheses. Log variables multiplied by 100 (approximate percent); rate and YoY variables in native units (percentage points). ** and * denote significance at the 90 and 68 percent confidence levels respectively. Peak is the largest absolute response across the full 0–60 month horizon; h^* is the horizon at which it occurs. HAC standard errors with Bartlett kernel, bandwidth $h + 1$.

	0	4	8	12	20	Peak	h^*
Real GDP (log)	-0.25*	-3.26**	-2.90**	-2.15**	-0.26	-3.61**	5
	(0.19)	(0.70)	(1.18)	(1.27)	(1.47)	(1.13)	
HH Consumption (log)	-0.74**	-2.49**	-3.42**	-2.81**	-0.51	-3.76**	5
	(0.23)	(0.68)	(1.48)	(1.46)	(1.77)	(1.48)	
HH Durables (log)	-1.44	-6.97**	-11.33**	-10.88**	4.79	-14.01**	10
	(1.55)	(3.34)	(4.45)	(5.06)	(5.65)	(5.56)	
Investment, GFCF (log)	-2.32**	-5.86**	-10.75**	-7.11**	-1.90	-10.75**	8
	(1.22)	(2.34)	(3.44)	(3.47)	(3.20)	(3.44)	
Exports (log)	-0.43	-4.54**	-1.55	2.01	5.13*	6.37**	17
	(1.28)	(1.41)	(1.68)	(2.36)	(3.23)	(3.15)	
Imports (log)	0.65	-5.15**	-7.20**	-5.98**	2.94*	-7.80**	9
	(0.96)	(2.13)	(2.44)	(3.18)	(2.25)	(2.66)	
Trade Balance / GDP	-0.07	0.38	1.61**	1.62*	-0.50	1.65**	9
	(0.31)	(0.58)	(0.81)	(0.99)	(0.75)	(0.94)	
Terms of Trade (log)	0.50	0.64	1.00*	-0.77*	-0.30	-2.17**	11
	(0.60)	(1.28)	(0.66)	(0.67)	(1.53)	(1.19)	
GDP Deflator (log)	0.02	1.48**	1.62*	1.36*	0.87	1.62*	8
	(0.41)	(0.82)	(1.00)	(1.29)	(2.13)	(1.00)	
Receipts / GDP	-0.18*	-0.57*	-1.00**	-0.67**	0.40**	-1.00**	8
	(0.14)	(0.52)	(0.44)	(0.27)	(0.22)	(0.44)	
Expenditure / GDP	0.01	0.93**	1.80**	1.28**	0.53	1.80**	8
	(0.06)	(0.38)	(0.71)	(0.73)	(0.57)	(0.71)	
Debt / GDP	0.10	3.04*	7.88**	9.37**	8.89**	10.21**	13
	(0.34)	(2.15)	(4.39)	(4.30)	(4.56)	(4.47)	

Table 6: *Quarterly Impulse Responses to a 25bp Fiscal News Shock.* Columns report point estimates at selected horizons (quarters) with standard errors in parentheses. Log variables multiplied by 100 (approximate percent); ratio variables in percentage points of GDP. ** and * denote significance at the 90 and 68 percent confidence levels respectively. Peak is the largest absolute response across the full 0–20 quarter horizon; h^* is the horizon at which it occurs. Fiscal flow variables are expressed as four-quarter trailing sums divided by four-quarter trailing GDP. HAC standard errors with Bartlett kernel, bandwidth $h + 1$.

	All quarters			Event quarters		
Predictor	F	p	N	F	p	N
Panel A: Daily — pre-announcement window (5 trading days)						
1Y Gilt Yield (Δ)	0.99	0.427	87		—	
3Y Gilt Yield (Δ)	0.81	0.548	87		—	
5Y Gilt Yield (Δ)	1.10	0.366	87		—	
10Y Gilt Yield (Δ)	1.42	0.227	87		—	
15Y Gilt Yield (Δ)	1.29	0.278	87		—	
GBP/USD (Δ)	1.48	0.206	86		—	
GBP/DEM (Δ)	0.70	0.625	86		—	
GBP ERI (Δ log)	0.85	0.521	87		—	
FTSE All-Share (Δ log)	1.24	0.299	86		—	
US 10Y Yield (Δ)	0.32	0.899	71		—	
Panel B: Quarterly — domestic macro (4 lags)						
Current Deficit / GDP	0.39	0.814	144	0.79	0.538	65
Total Expenditure / GDP	1.02	0.399	144	1.62	0.180	65
Δ Receipts / GDP	0.41	0.801	144	0.22	0.925	65
Δ Debt / GDP	0.15	0.964	144	0.41	0.799	65
Δ HH Consumption (log)	0.46	0.764	144	0.44	0.777	65
Δ Government Consumption (log)	0.91	0.460	144	0.72	0.580	65
Δ Exports (log)	1.44	0.223	144	0.47	0.758	65
Δ Imports (log)	2.30	0.062*	144	1.63	0.178	65
Δ Trade Balance / GDP	1.08	0.370	144	0.89	0.474	65
Δ Employee Compensation	0.66	0.618	144	1.31	0.275	65
Δ Investment, GFCF (log)	0.53	0.711	144	0.06	0.993	65
Δ GDP Deflator (log)	1.58	0.184	144	1.26	0.297	65
Panel C: Quarterly — global macro (4 lags)						
Δ Real WTI Oil Price (log)	1.52	0.199	144	0.46	0.762	65

Table 7: Predictability of Fiscal News Shocks, 1980-2019

C. Data: Sources and Calculations

This section outlines how we build our dataset. Primary sources are listed. Where we have modified the raw data, calculation is included. Full details can be found in the online Python notebook which compiles and constructs all relevant data using online sources only. In the interest of transparency and reproducibility, the entire paper is contained on one replication file.

C.1 Calculations

The following calculations and transformations were used:

- **GBP ERI** Nominal effective exchange rate (NEER) index for sterling, trade-weighted. The benchmark series is sourced from Refinitiv-Eikon. For the period prior to the Refinitiv coverage, the series is extended using the Bank of England sterling exchange rate index (XUDLBK67), rescaled to match the level of the Refinitiv series at the splice date using a multiplicative scaling factor. The splice quality is validated over the common sample period 1990–2018 ($n = 7{,}076$ daily observations): the correlation of levels is 0.9998 and the correlation of daily log returns is 0.9960, confirming near-perfect agreement between the two series. Entered as log level. XUDLBK67.

- **The Consumer Price Index** is calculated by the ONS from 1988m1 onwards. To extend consumer prices back to 1980, we splice using the CPI Historical Series (extending the CPI methodology backward from 1988), using growth rates to extend back in time. For CPI subindices we do not extend due to lack of closely related subindices.

- **Variables expressed as a percentage of GDP** are annualised and computed as the four-quarter trailing-sum of the outcome divided by the four-quarter trailing sum of GDP - they are the cumulative outcome over the past year over GDP for the past year. This is to strip out excessive seasonal variation in ratios where both numerator and denominator will experience quarter-to-quarter seasonality.

Variable	Definition & Transformation	Ticker	Source
Fiscal surprise (s_t)	Change in the 3-year gilt yield on UK Budget days, zero on all other days.	—	Bank of England nominal yield curve
Gilt yields, $y_t^{(n)}$	Benchmark spot yields on UK government bonds at constant maturities of 1–20 years. Complete coverage for maturities 1–16 years; longer maturities are less reliable due to infrequent issuance. Entered in percentage points.	—	Bank of England, nominal yield curve
Inflation breakevens	Implied inflation expectations derived from the spread between nominal and index-linked gilt spot yields at maturities of 3, 5, and 10 years. No shorter horizon is available.	—	Bank of England, inflation curve (from 1985)
Term Premium	Decomposition of the 1,3,5,10-year nominal yield into expected short rate and term premium components following Adrian, Crump & Moench (2013), estimated on the BoE yield curve.	—	Authors' calculations using Bank of England yield curve
Bilateral exchange rates	Sterling–US dollar (GBP/USD) and sterling–euro (GBP/EUR) spot rates. For the pre-euro period (before 1999) the GBP/EUR series is constructed from the GBP/DEM rate, rebased using the fixed DEM/EUR conversion rate of 1.95583.	GBP=, EURGBP=	Refinitiv-Eikon
GBP effective exchange rate	Nominal effective exchange rate (NEER) index for sterling, trade-weighted. The benchmark series is sourced from Refinitiv-Eikon; for periods of limited coverage, the series is spliced with the Bank of England sterling exchange rate index. The splice is performed by matching levels at the overlap date; the correlation of daily returns in the overlap period exceeds 0.99.	—	Bank of England; Refinitiv-Eikon
Equity price index	FTSE All-Share index (price return index). Entered in logs.	.FTAS	Refinitiv-Eikon
Equity total return index	FTSE All-Share total return index, which reinvests dividends. Entered in logs. Preferred to the price index for longer-horizon analysis.	.FTASF	Refinitiv-Eikon

Table 8: Financial market variables. Daily frequency unless otherwise stated. Sample: 1980–2025.

Variable	Definition & Transformation	Ticker	Source
Industrial production (Mfg)	Index of UK industrial production (manufacturing, i.e. excludes Mining & quarrying (including North Sea oil & gas) Electricity, gas & water supply). Entered as log level; the dependent variable in monthly local projections is the cumulative log change $\log y_{t+h} - \log y_{t-1}$.	K22A	ONS DIOP dataset
Unemployment rate	UK Labour Force Survey (LFS) unemployment rate, seasonally adjusted. Entered in percentage points.	MGSX	ONS
CPI inflation	Consumer price index, year-on-year percentage change. For the pre-CPI period (before 1988) the series is spliced with the Retail Price Index (RPI) by matching levels at the overlap date.	D7G7	ONS; authors' calculations
Producer Inputs inflation	Producer price index for imported materials and fuels, year-on-year log change. Used to identify the import price pass-through channel of fiscal transmission via the exchange rate.	—	ONS PPI (from 1985)
CPI components	Sub-indices of the CPI including goods, services, energy, and food components. Year-on-year percentage changes. Used in the inflation decomposition analysis.	—	ONS (from 1988)

Table 9: Macroeconomic variables, monthly frequency. Sample: 1980–2019 (baseline); extended to 2025 for robustness.

Variable	Definition & Transformation	Ticker	Source
Real GDP	Gross domestic product, chained volume measure, seasonally adjusted. Entered as log level.	ABMI	ONS Quarterly National Accounts
Real GDP per capita	Real GDP divided by UK resident population (mid-year estimates, quarterly interpolation). Entered as log level.	IHXW, EBAQ	ONS QNA
Consumption	Household final consumption expenditure, chained volume measure, seasonally adjusted. Entered as log level. Decomposed into durables.	ABJR, UTID, UTIT, UTIL, UTIP	ONS QNA
Government cons.	General government final consumption expenditure, chained volume measure, seasonally adjusted. Entered as log level.	NMRY	ONS QNA
Investment (GFCF)	Gross fixed capital formation, total economy, chained volume measure, seasonally adjusted. Entered as log level.	NPQT	ONS QNA
Exports & imports	Total trade in goods and services, chained volume measure, seasonally adjusted. Entered as log levels.	IKBK, IKBL	ONS QNA
Terms of trade	Ratio of the export goods price deflator to the import goods price deflator, both constructed as the ratio of current price to chained volume measure series for total goods trade. Entered as log.	BOKG, BOKH, BQKQ, BQKO	ONS QNA; authors' calculations
GDP deflator	Implied deflator for GDP at market prices, seasonally adjusted.	YBGB	ONS QNA

Table 10: Macroeconomic variables, quarterly frequency. Sample: 1980–2019 (baseline); extended to 2025 for robustness. All log-level variables enter the local projection as long-difference $y_{t+h} - y_{t-1}$.